Section 2

Electrophysiology & Physiological Properties of Cardiac Muscle

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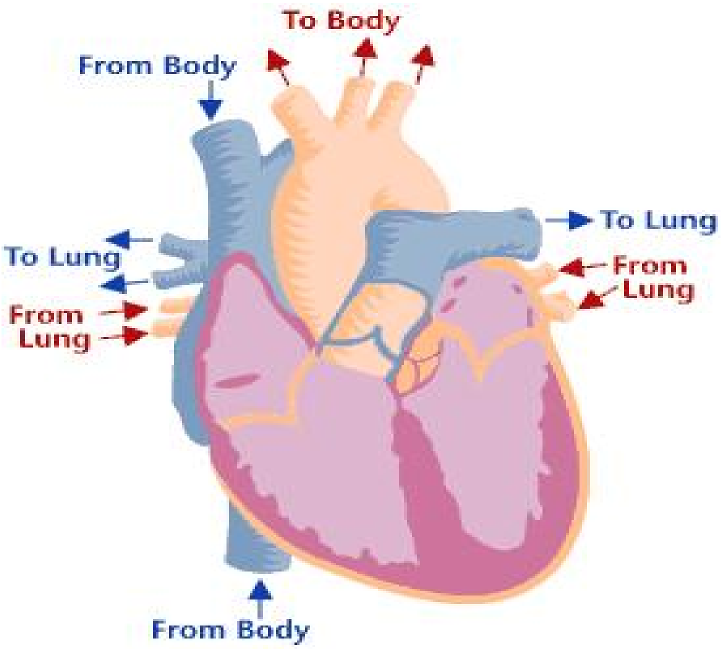
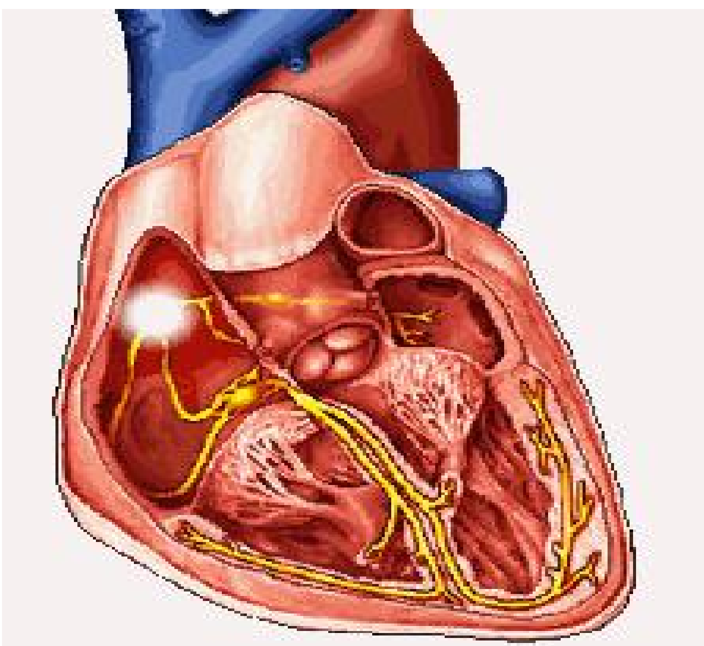
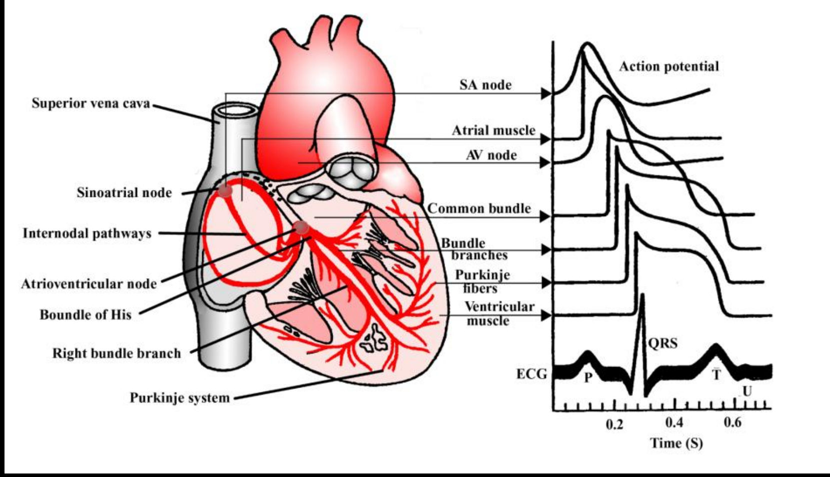




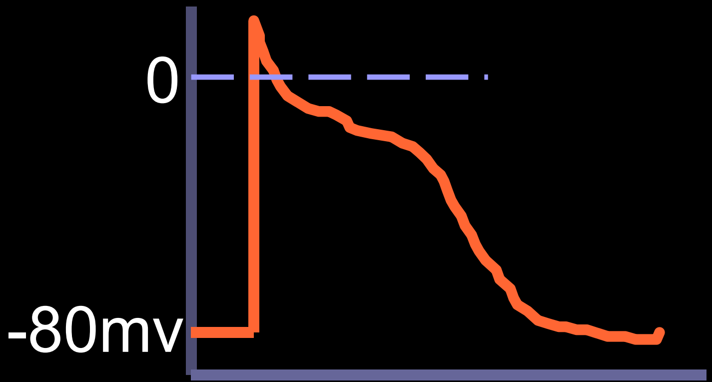
- Cardiac AP starts in SA Node, propagates in an orderly fashion throughout the heart

- A wave of depolarization that sweeps over the heart from its base to its apex and from the endocardial to the epicardial surface.

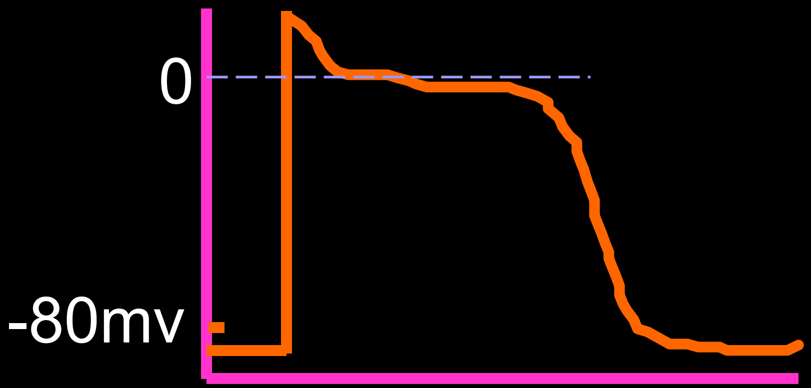
Action potentials from different areas of the heart



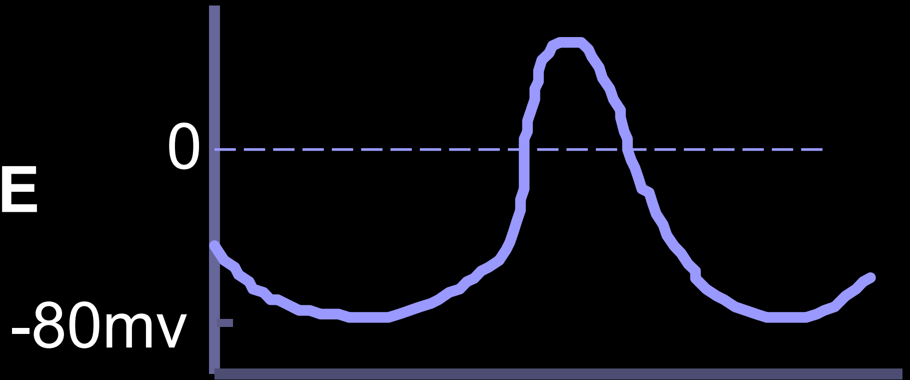
ATRIUM



VENTRICLE

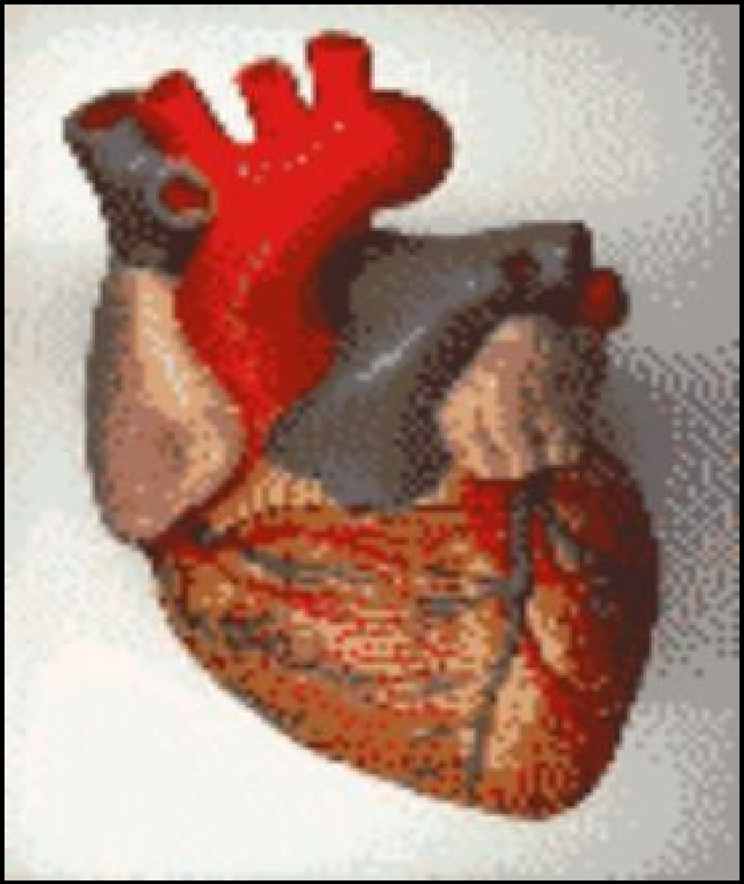


SA NODE



time

Two kinds of cardiac cells



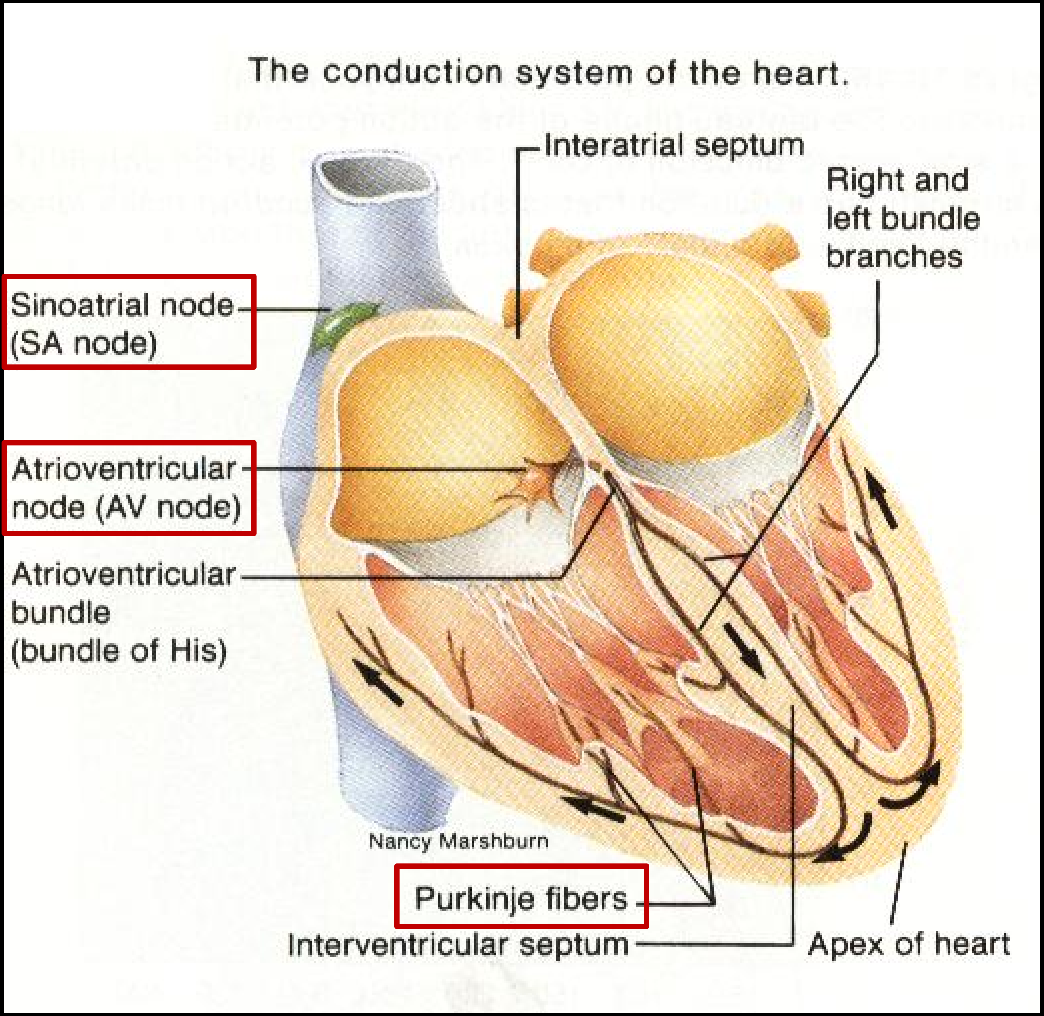
1, The working cells.

- ✓ Atrial muscle
- ✓ Ventricular muscle

Special property: contractility

2, Autorhythmic cells system

Special property: autorhythmicity



- 工作细胞
  - 心房肌
  - 心室肌

*有稳定的静息电位
*主要执行收缩功能

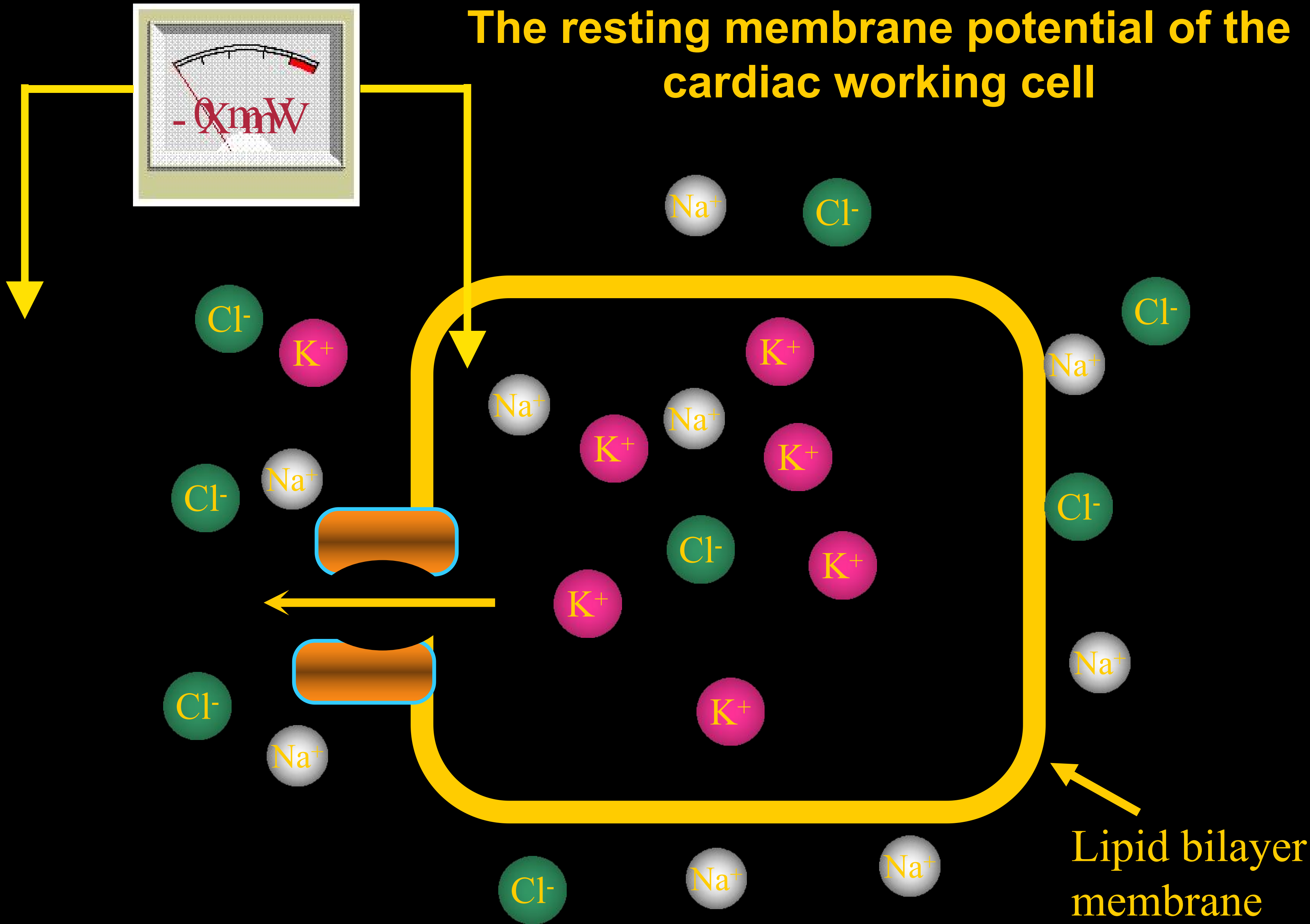
- 自律细胞
  - 浦肯野
  - 窦房结
  - 房室结

快反应细胞：心房肌、心室肌、浦肯野

慢反应细胞：窦房结、房室结

I. Transmembrane Potentials of Myocardial Working Cells

The resting membrane potential of the cardiac working cell



The resting membrane potential of the cardiac working cell (as the other cells)

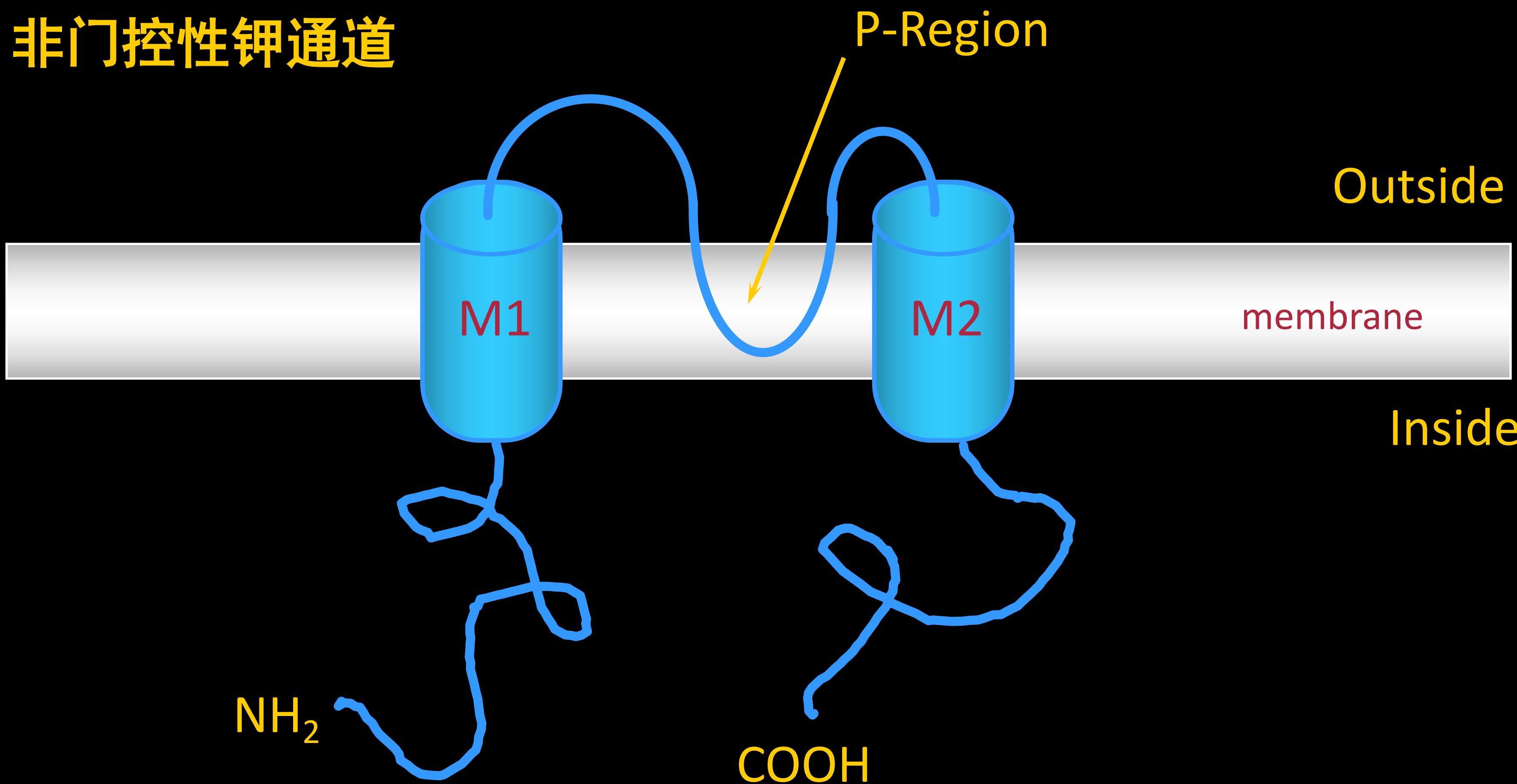
- ✓ -90 ~ -80 mV
- ✓ K^+ Equilibrium potential I_{K1} 通道(内向整流钾通道)
- ✓ Permeability (Na^+ background current)
- ✓ Activity of Na^+- K^+ pump (pump current, I_{pump})

Inward Rectifier (I_{k1}) Channel

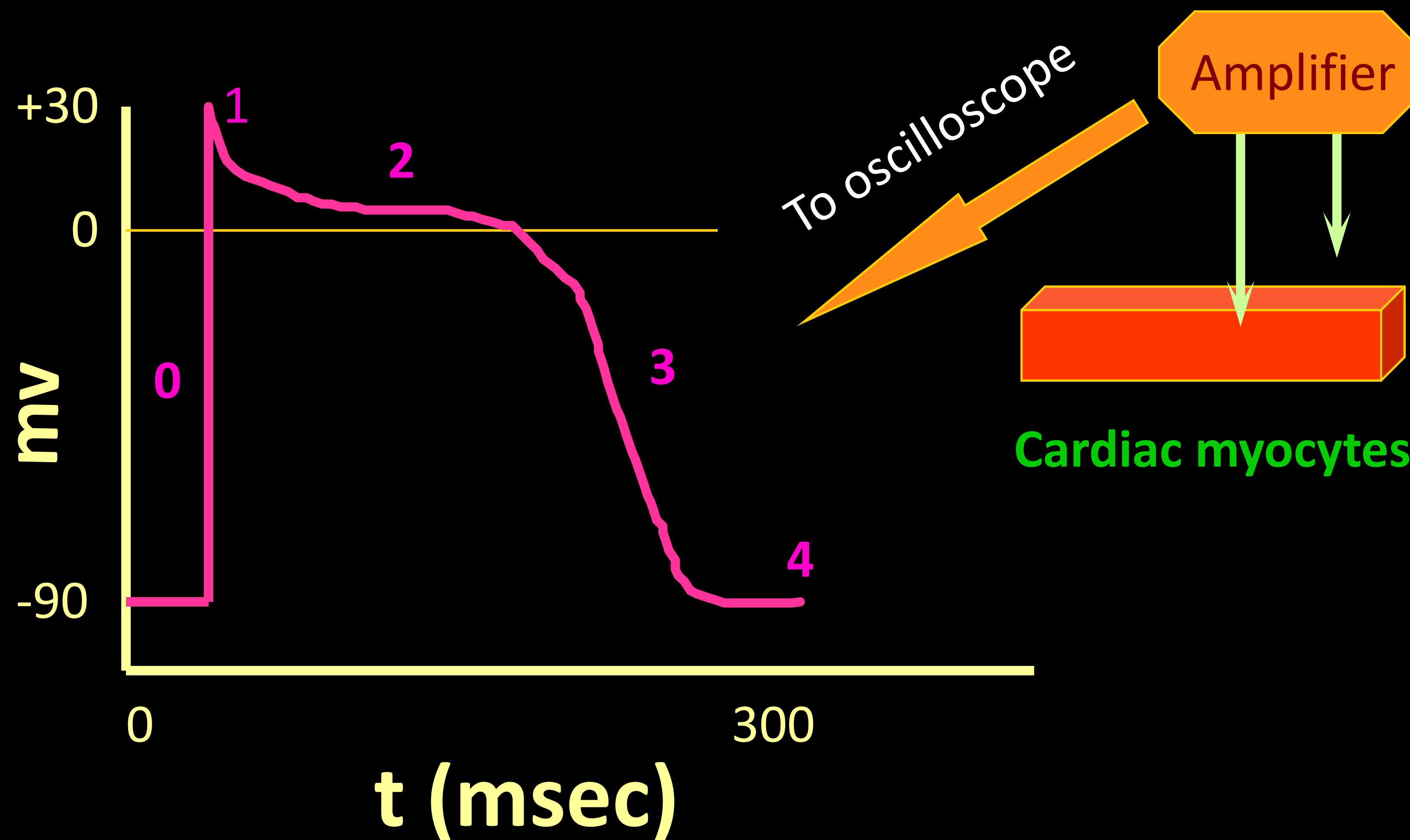
内向整流钾通道

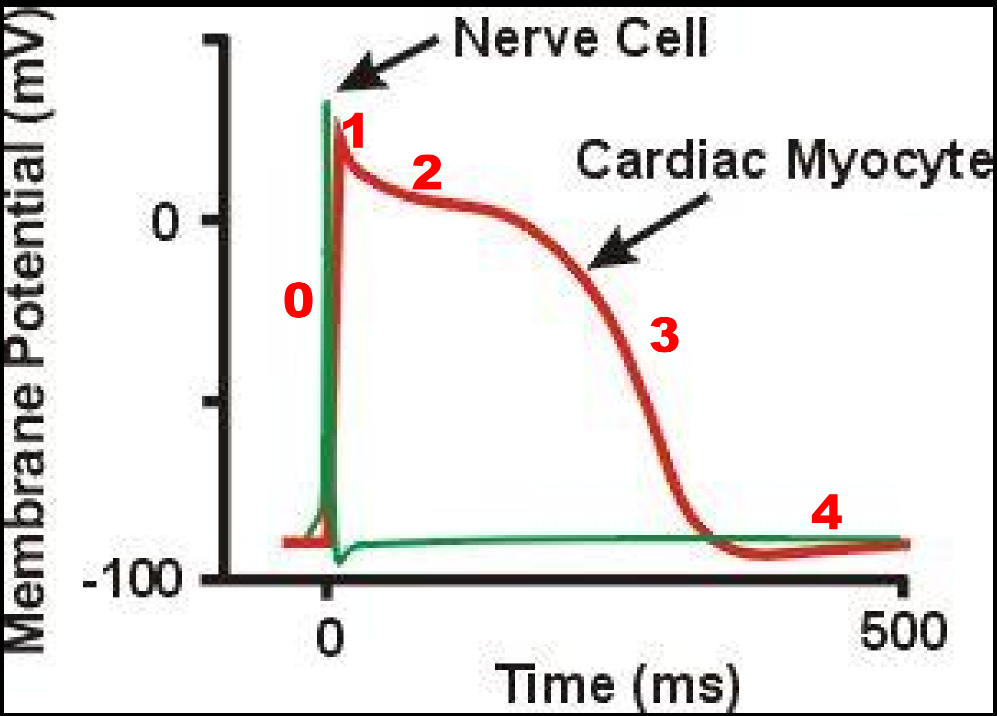
Not “voltage/chemical sensor”

非门控性钾通道



Electrophysiology of the ventricular myocytes





Why is the action potential of cardiac muscle so long?

Phase 0 (快速去极化):
rapid depolarization, 1-2ms

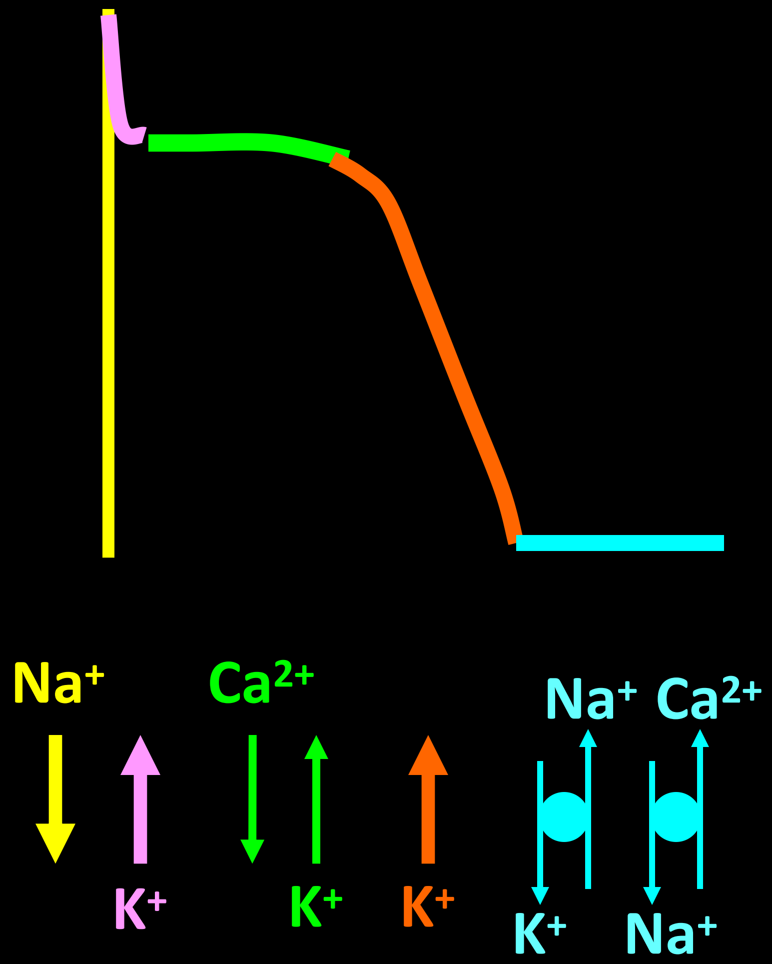
Phase 1（快速复极化初期）：
early rapid repolarization, 10 ms

Phase 2（平台期）: plateau,
slow repolarization,
the potential is around 0 mv.
100 – 150ms

Phase 3（快速复极化末期）：
late rapid repolarization.
100 – 150 ms

Phase 4 （静息期）：
resting potentials

Ion Currents in Myocardial Working Cells



0期：Na^+快速内流

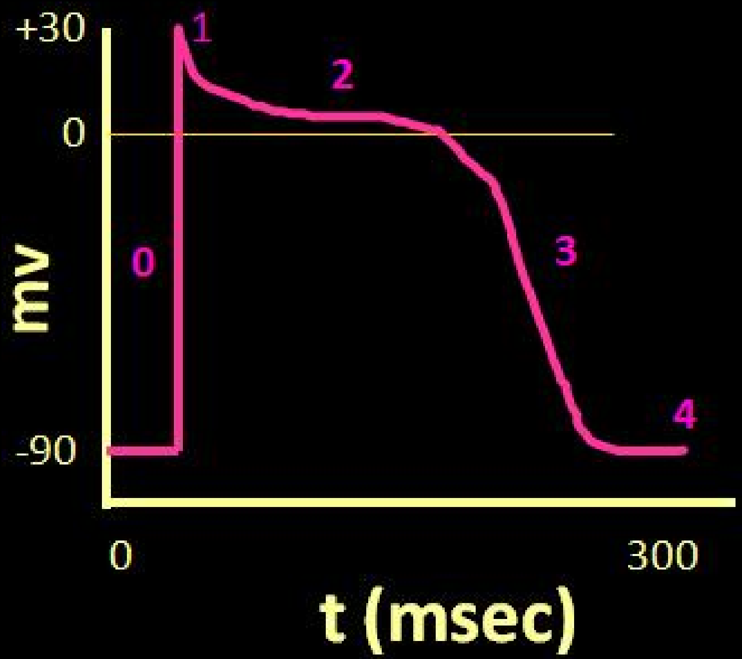
1期：K^+ 外流

2期：Ca^{2+}内流 达平衡
K^+ 外流

3期：K^+ 外流

4期：Ca^{2+}—Na^+ 交换
Na^+—K^+交换(钠泵)

Phase 0:
rapid depolarization,
-90 ~ +30 mv



✓过程短暂 (1~2ms)

✓幅度大 (120mv)

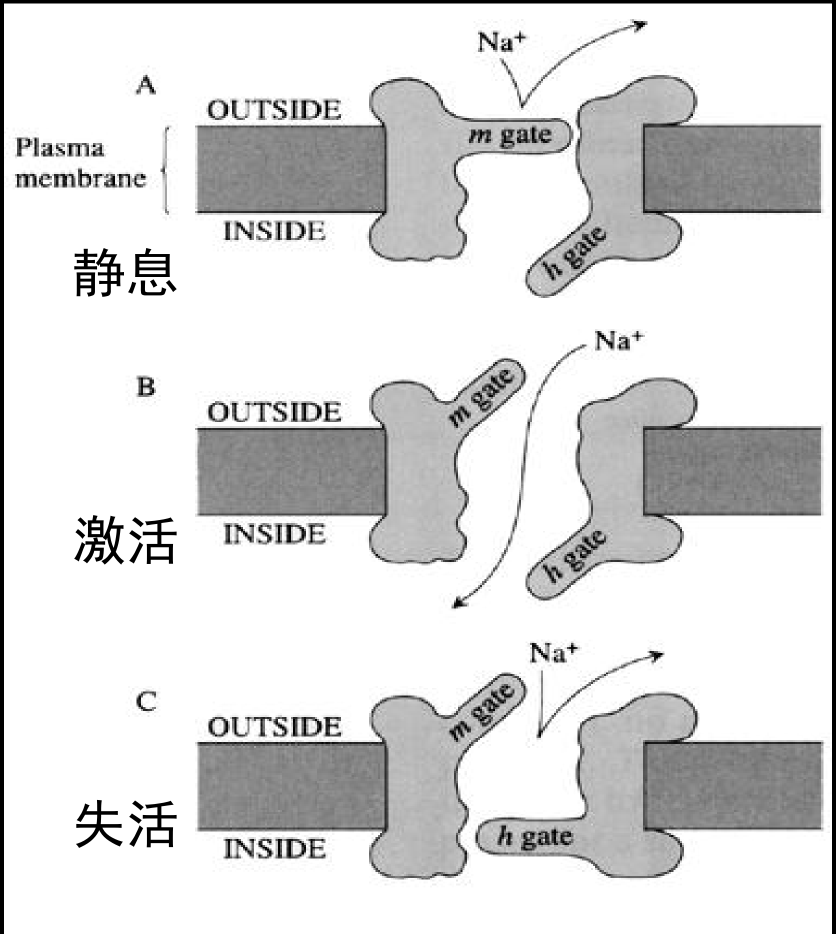
✓速度快 (200~400v/s)

Cardiac Na^+ Channels

- ✓ 快钠通道 ← I 类抗心律失常药(奎尼丁)
- 快激活 & 快失活

→AP in fast response cells
快反应细胞

- ✓ Almost identical to nerve Na^+ channels (structurally and functionally)
- TTX sensitivity 1/100-1/1000



Phase 1:
Early rapid
Repolarization,
+30 ~ 0mv
(10 ms)

Fast Na^+ channels closed

Fast Na^+ current inactivation

↓

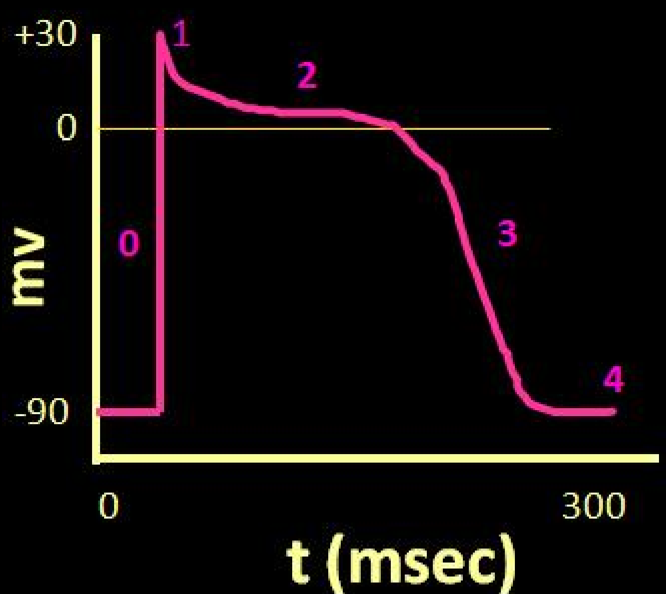
K^+ channel open

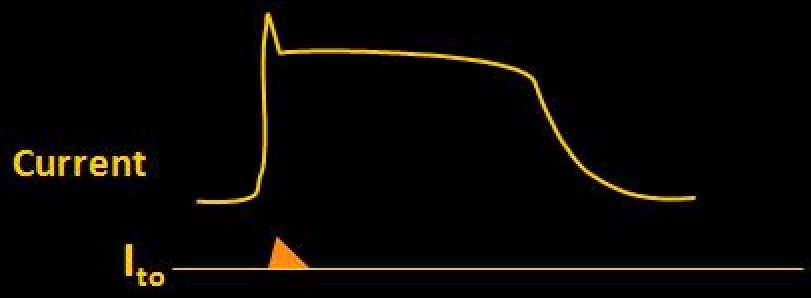
↓

transient K^+ outward current

瞬时外向电流（I_{to}）↖

4-氨基吡啶





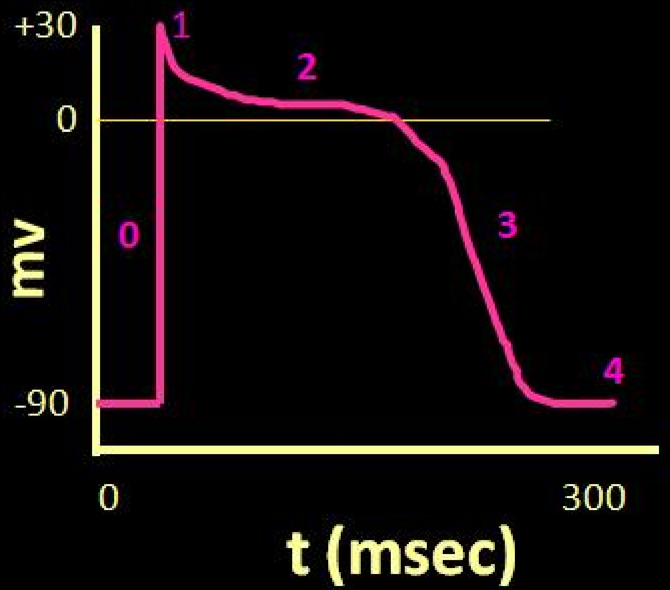
Phase 2:
Prolonged plateau,
Around 0mv
(100-150 ms)

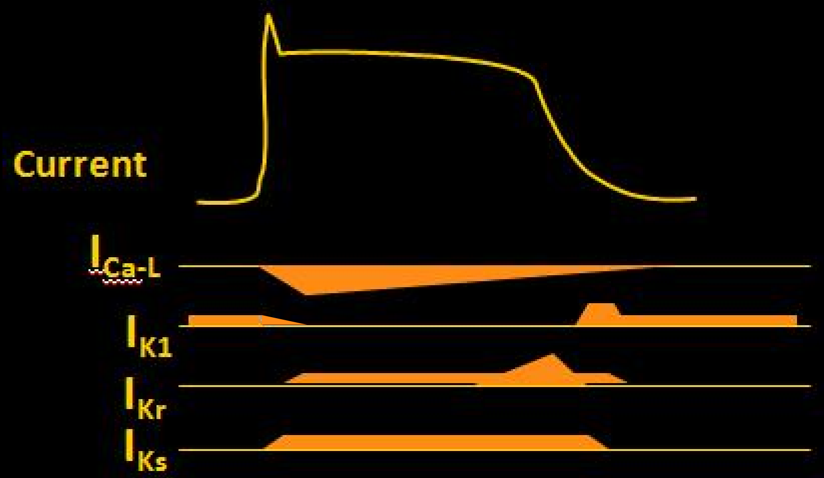
Outward current: K^+

- ✓ I_{K1} (内向整流钾电流) ↓
- ✓ I_K (延迟整流钾电流)
- ✓ I_{pump}

Inward current: Ca^{2+}

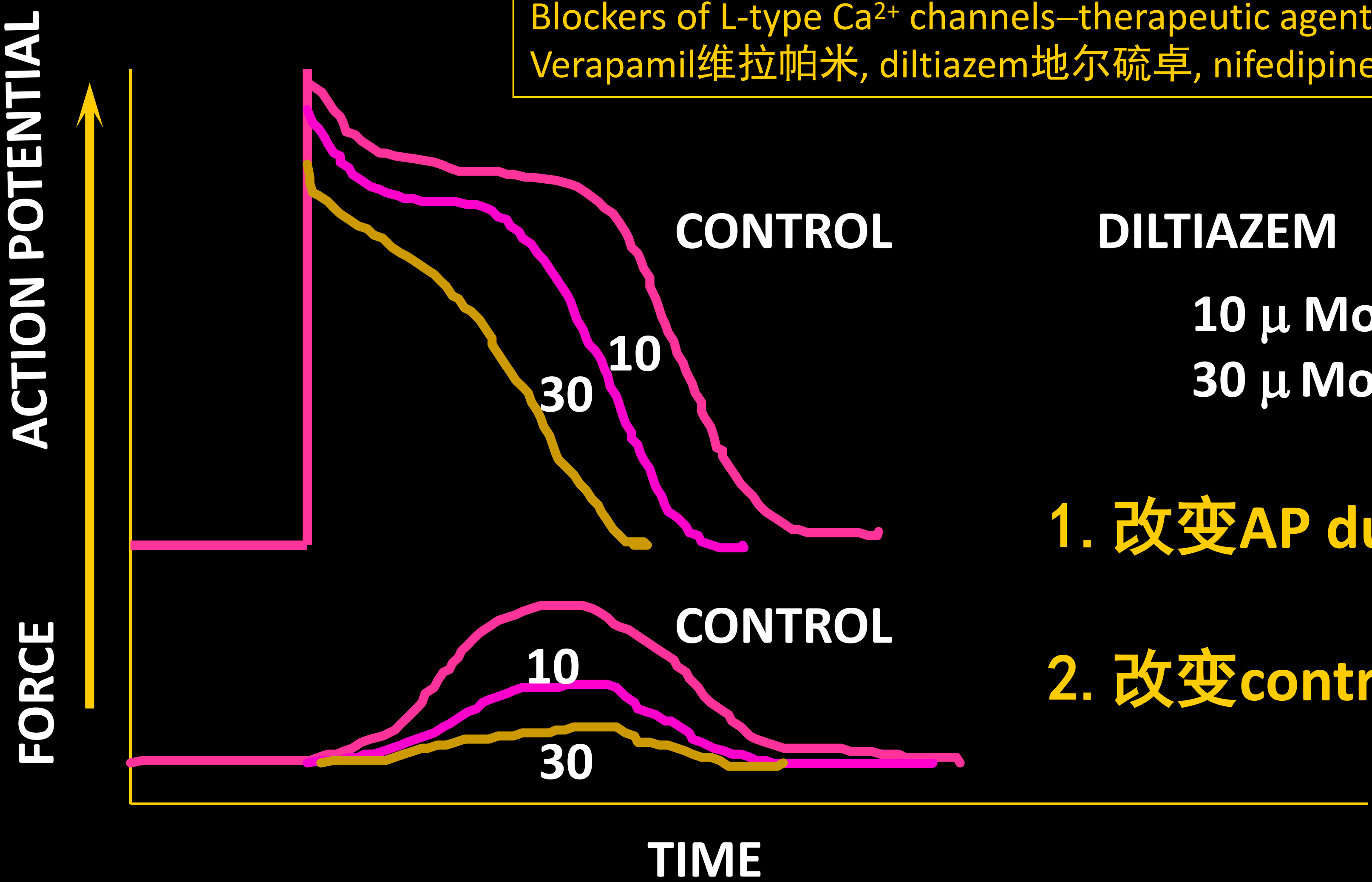
- ✓ $I_{ca\text{-}L}$ (L- Ca^{2+} channels) ↖ 维拉帕米
- ✓ I_{Na} (慢失活)
- ✓ $I_{Na\text{-}Ca}$





Ca^{2+} channel controls action potential duration & contractility

Blockers of L-type Ca^{2+} channels–therapeutic agents:
Verapamil维拉帕米, diltiazem地尔硫卓, nifedipine硝苯地平



DILTIAZEM

10 μ Mol/L
30 μ Mol/L

1. 改变AP duration

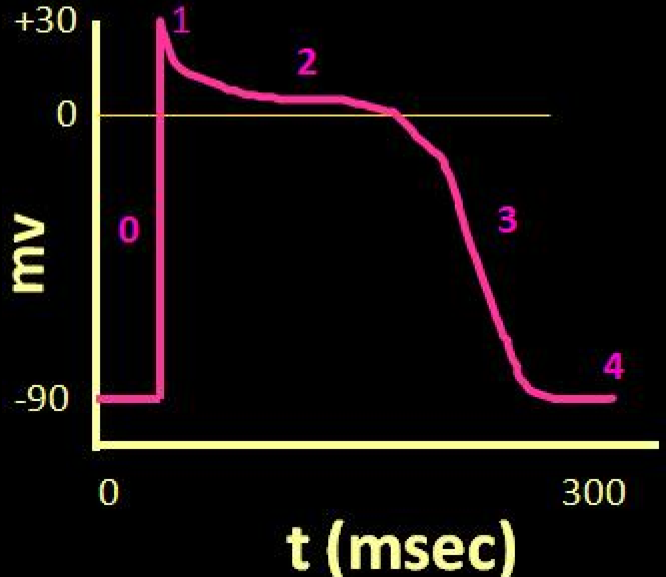
2. 改变contractility

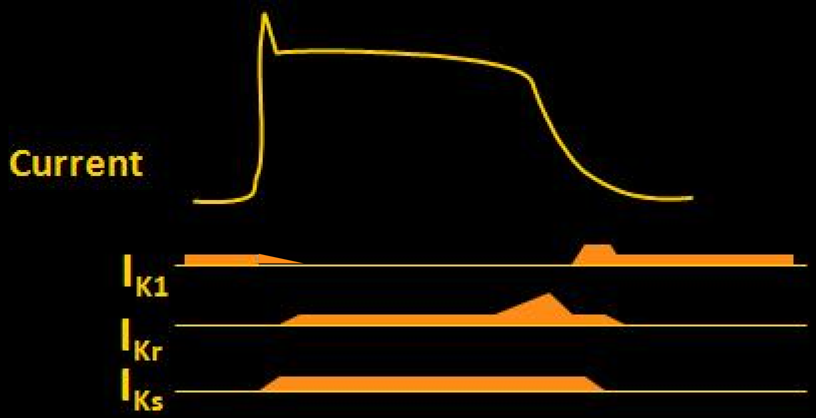
Phase 3:
late rapid repolarization,
(100-150 ms)

Outward current:

- ✓ I_{K1} (内向整流钾电流)
- ✓ I_K (延迟整流钾电流)
- ✓ 其他

III 类抗心律失常药





Cardiac K^+ Channels

- □ 内向整流钾通道（I_{K1}）

本质上是非门控性钾通道

- ✓ Sets resting potential in cardiac working cells (通透性高)
- ✓ Contributes to phase 2 plateau (通透性低)
- ✓ Contributes to phase 3 repolarization (通透性高)

Cardiac K^+ Channels

- ☐ 介导瞬间外向电流的钾通道（I_{to}）
- ☐ 延迟整流钾通道（I_K）

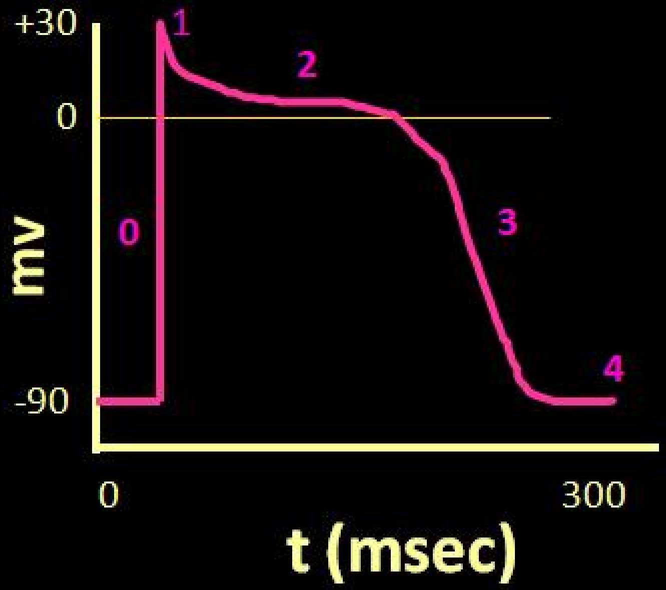
本质上都是电压门控性钾通道
（神经细胞复极 – I_K）

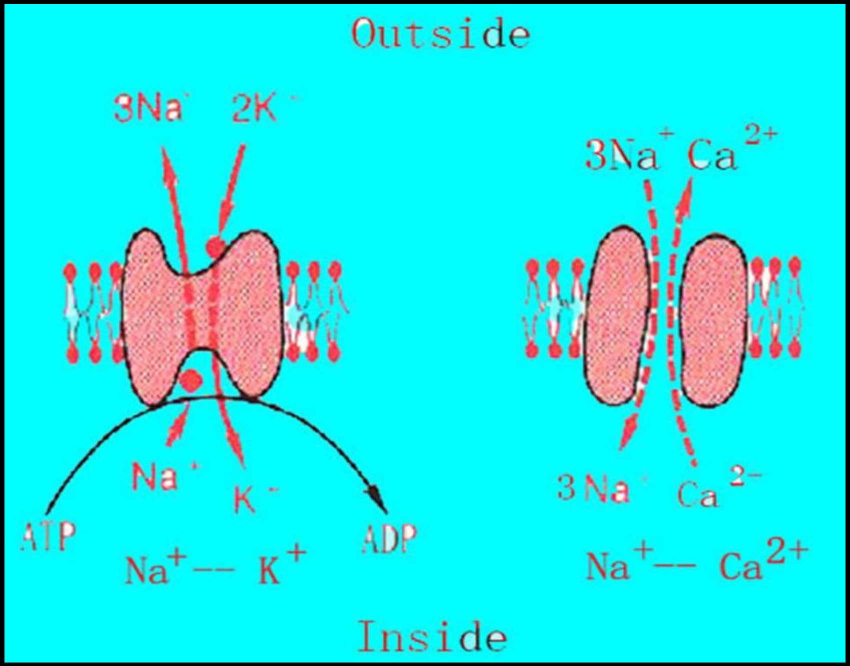
- ➢ I_{to} - phase 1 rapid repolarization
- ➢ I_K - fights with Ca^{2+} to maintain plateau in phase 2
- ➢ I_K - Contributes to phase 3 repolarization

Phase 4:
resting potentials

Concentration gradient of ions is recovered

- ✓**Take out intracellular Na^+ & Ca^{2+}**
- ✓**Intaking back K^+**



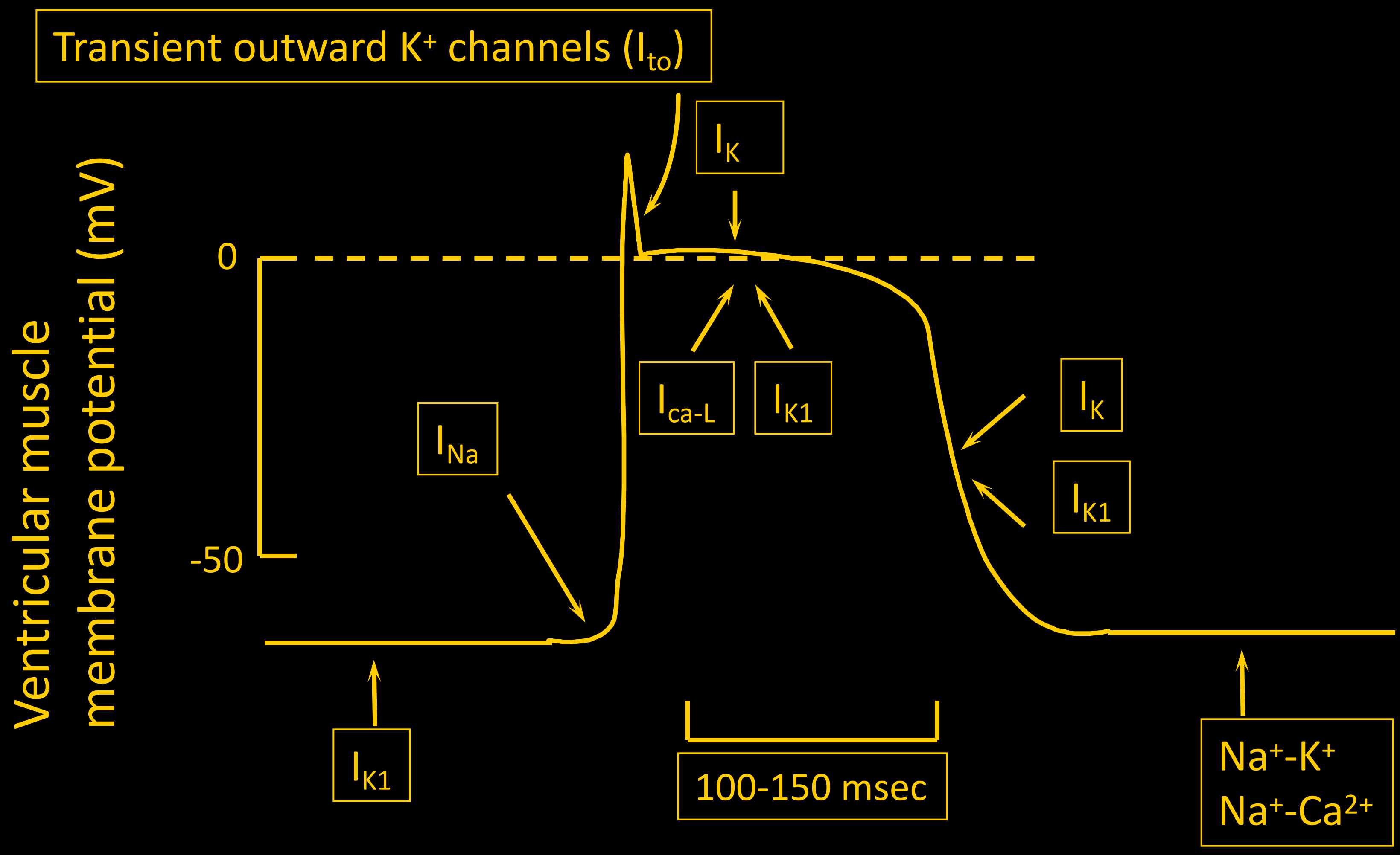


Summary of Action Potential of a Myocardial Contractile Cell

- Depolarization= Sodium Influx
- Early Rapid Repolarization= Potassium Efflux
- Plateau= Calcium Influx & Potassium Efflux
- Late Rapid Repolarization= Potassium Efflux

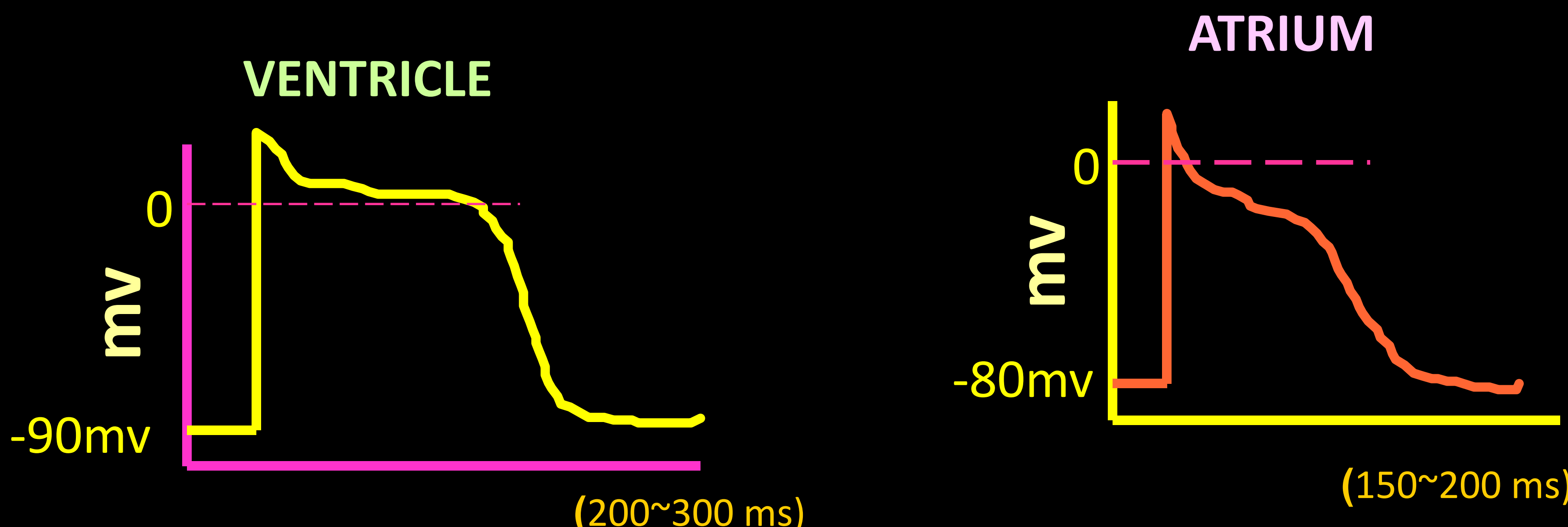
从0期去极化开始到3期复极化完毕这段时间，称为动作电位时程（Action potential duration, APD） 200~300 ms

Ion Channels in Ventricular Myocytes

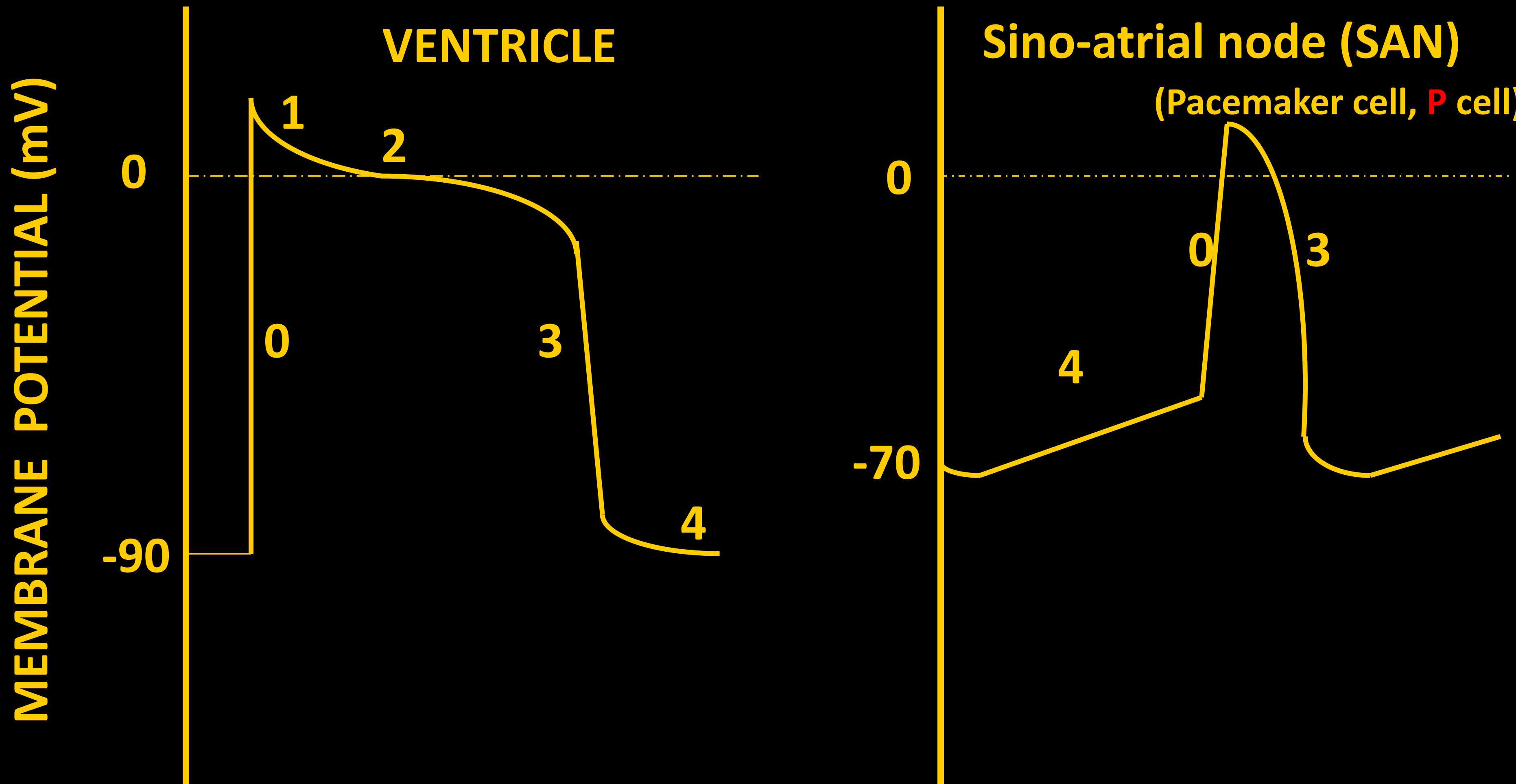


Ion Channels in Atrial Cells

- ✓ I_{K1} 密度稍低，静息电位-80mv
- ✓ I_{to}较发达，持续到2期，平台期不明显，复极化快，动作电位时程短
- ✓ 乙酰胆碱敏感的钾通道$I_{K\text{-}Ach}$，Ach作用下K^+外流增强，动作电位时程缩短



II. Transmembrane Potentials of Myocardial Rhythmic Cells



- ✓ 最大复极电位(Maximal repolarization potential, MRP)-70mv
- ✓ 动作电位去极化的速度和幅度均较小
- ✓ No phase 1 and 2
- ✓ 4期自动去极化，到达阈电位时便可爆发AP

Maximum repolarization potential

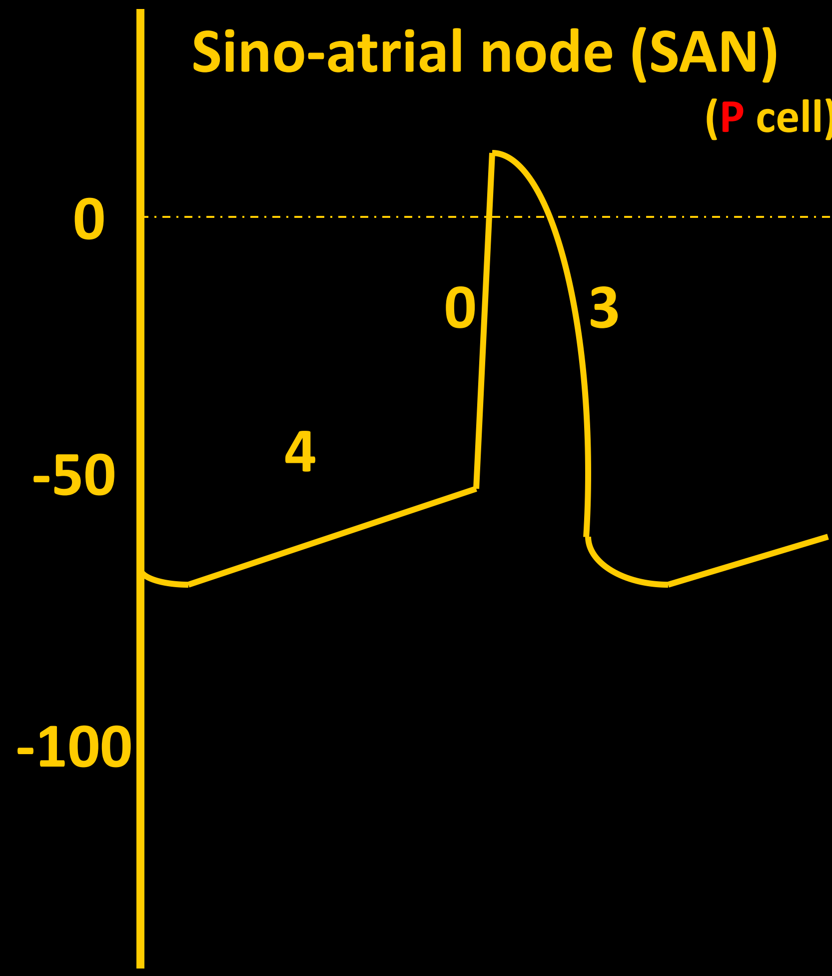
(最大复极电位)

自律性细胞动作电位3期复极末达到最大极化状态时的电位值代表静息电位

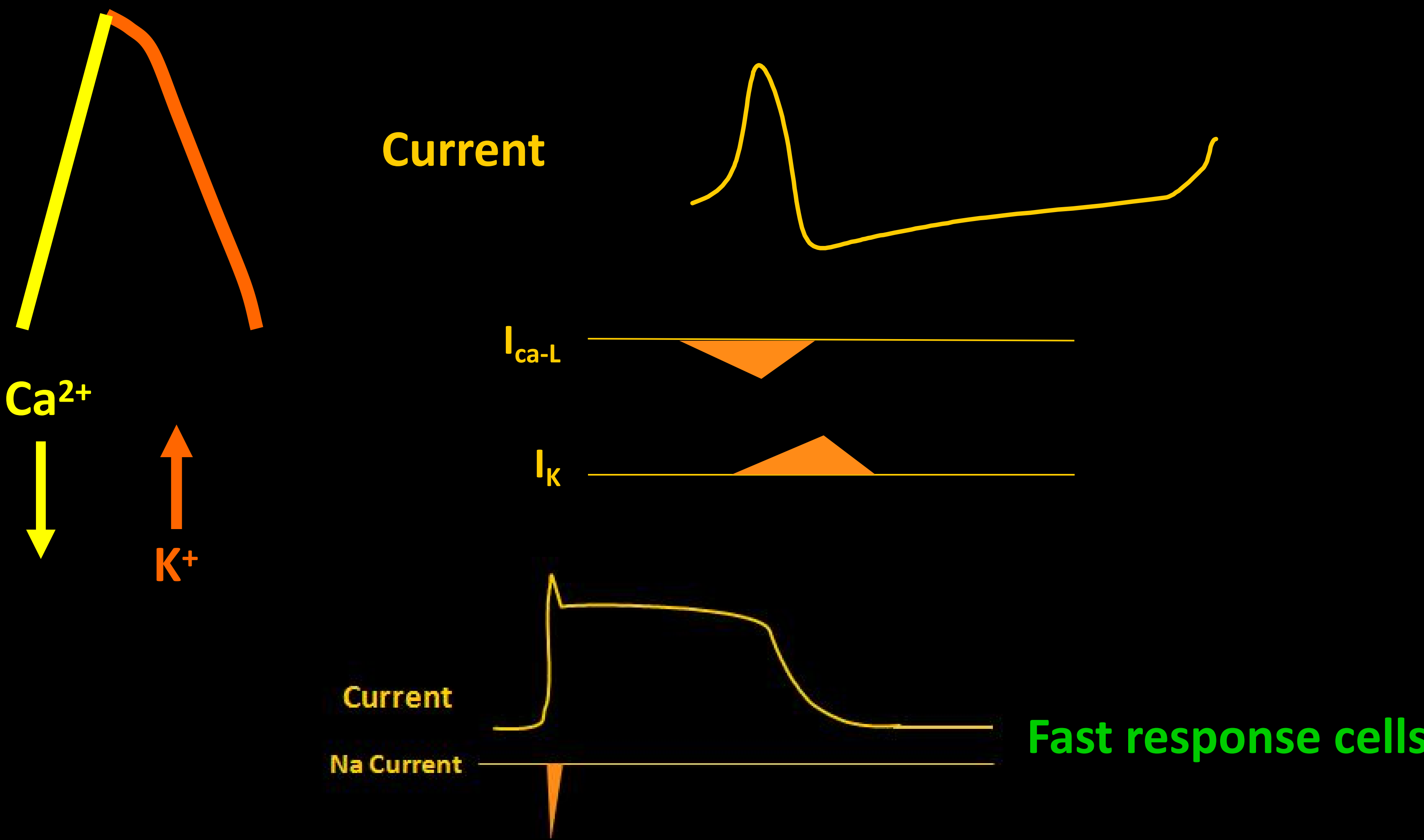
Phase 4 spontaneous depolarization

(4期自动去极化)

4期膜电位并不稳定，而是立刻开始自动去极化



SA Node Cells Slow response cell慢反应细胞



Cardiac Ca^{2+} Channels

□ L-type   L: long-lasting

↖ 维拉帕米

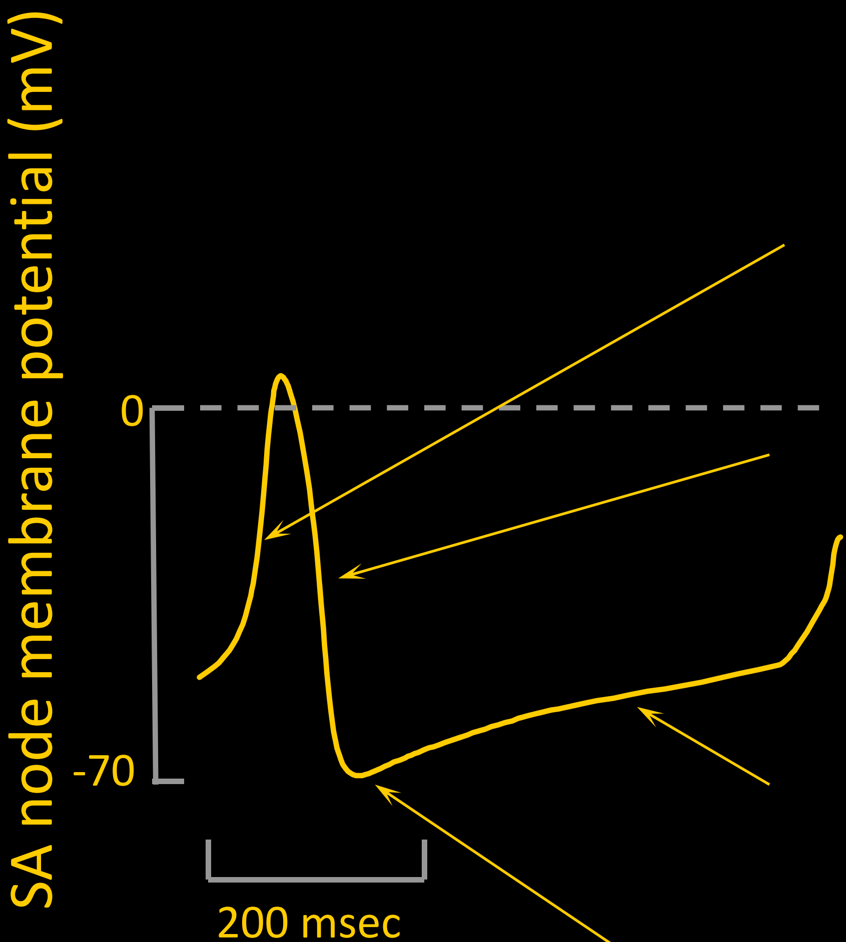
✓ Structurally similar to Na^{+} channels

(两个闸门/三种功能状态)

✓ Functionally different from Na^{+} channels

- 慢激活(去极化速度慢，去极化幅度小)
- 慢失活(持续时间长)

SA Node Action Potential



Voltage-gated Ca^{2+} channels($I_{ca\text{-}L}$)

No I_{Na} channels

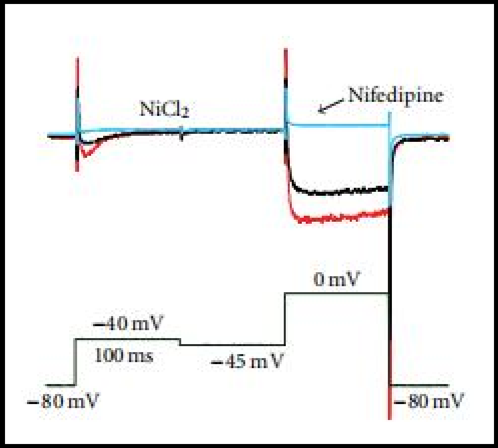
Voltage-gated K^{+} channels (I_{K})

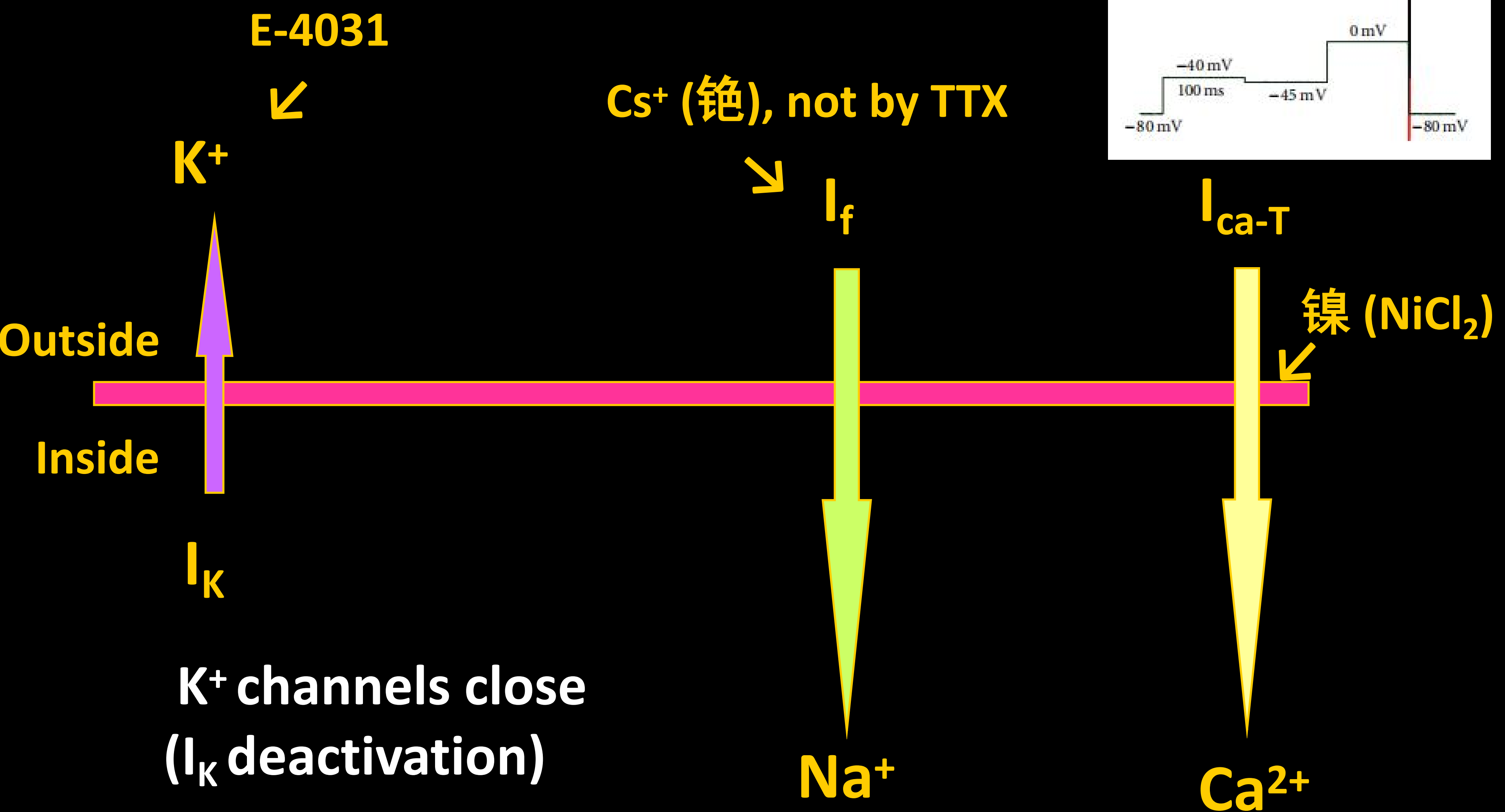
No I_{to} channels

Pacemaker currents

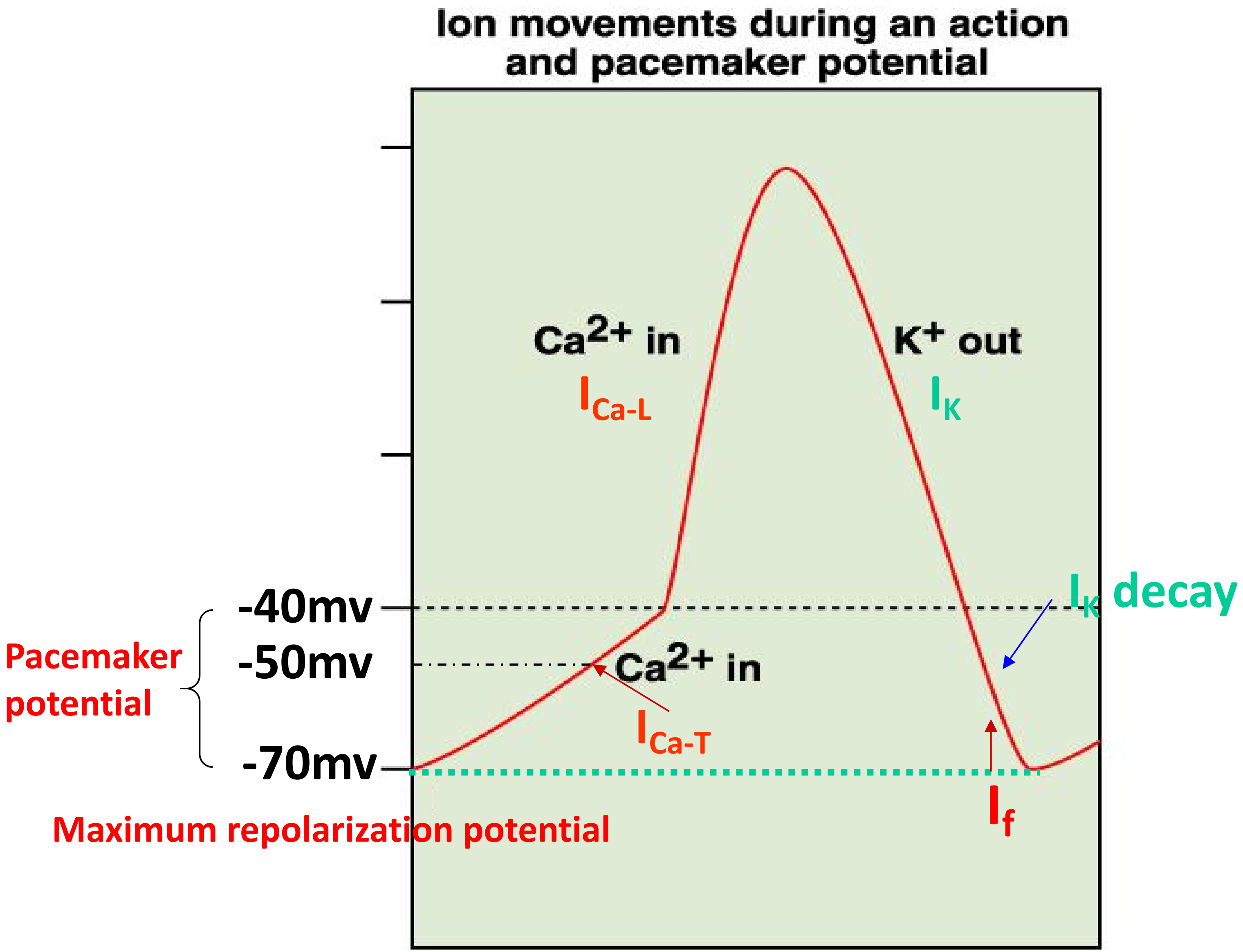
Less I_{k1} channels

起搏电流Pacemaker current





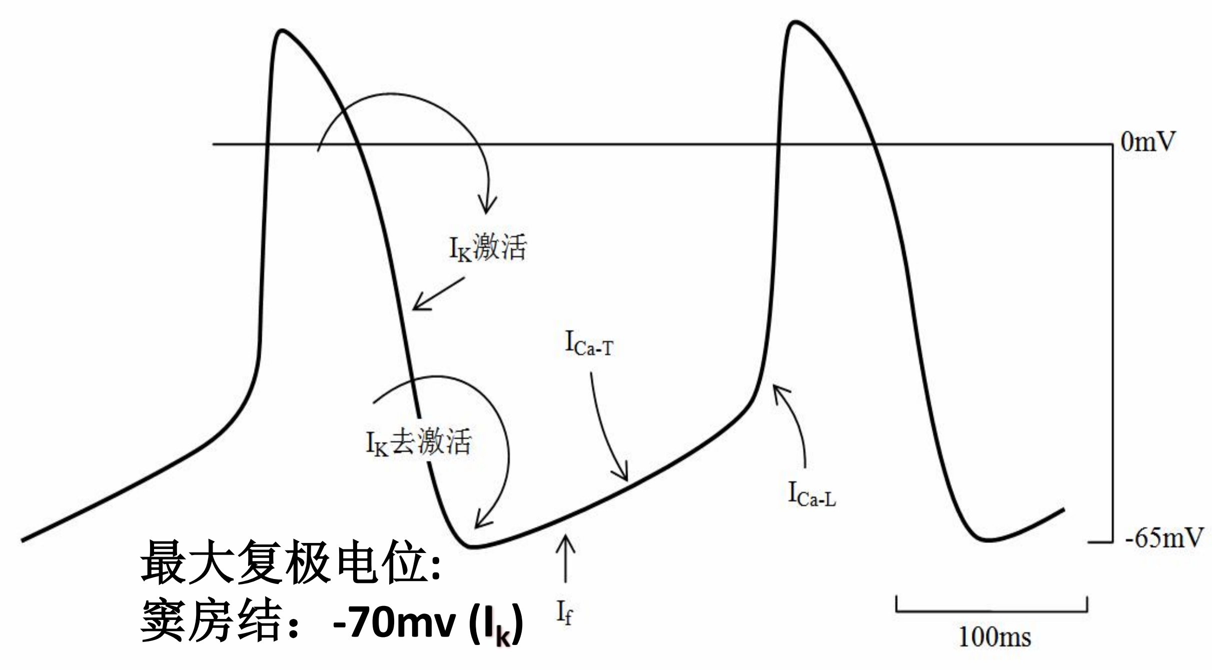
Pacemaker activity: spontaneous time-dependent depolarization that leads to an AP during Phase 4



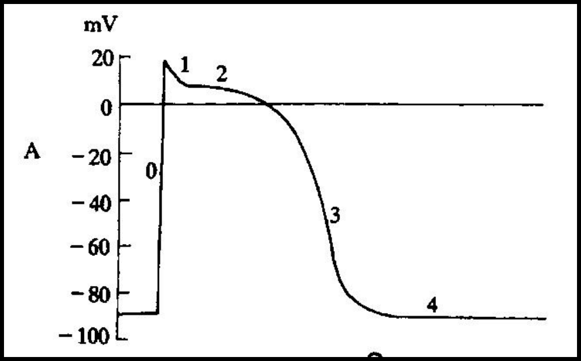
I_f Current

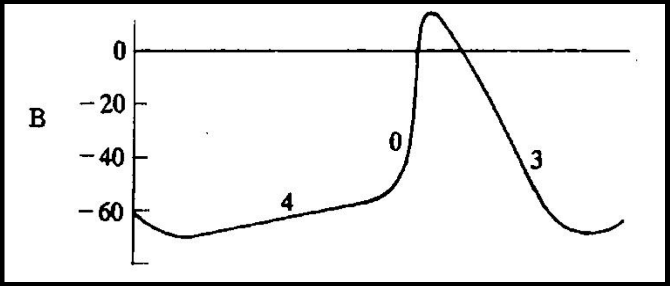
超极化激活的内向离子电流 **f: funny**

- 非选择性阳离子通道，超极化激活
- 3期复极化至-50mv时，I_k通道开始关闭，电流逐渐减小，同时 I_f 通道开始缓慢激活开放
- I_f在-100mv时才充分激活(P细胞MRP只有-70mv)，故不如I_k重要 (I_f与I_k作用之比为1:6)



Phase \ Type	Fast response cell (atria/ventricle)	Slow response cells (SA/AV node)
Phase 0	I_{Na} Channels open	$I_{Ca\text{-}L}$ channels open
Phase 1	I_{to} (K^+) Channels open	No
Phase 2 (Plateau)	$I_{Ca\text{-}L}$ channels open is balanced against efflux of I_k	No
Phase 3	I_K, I_{K1} channels open	I_K channels open
Phase 4	Steady	Spontaneously depolarize, I_K decay; I_f & $I_{Ca\text{-}T}$ channels open





Purkinje cells/fibers

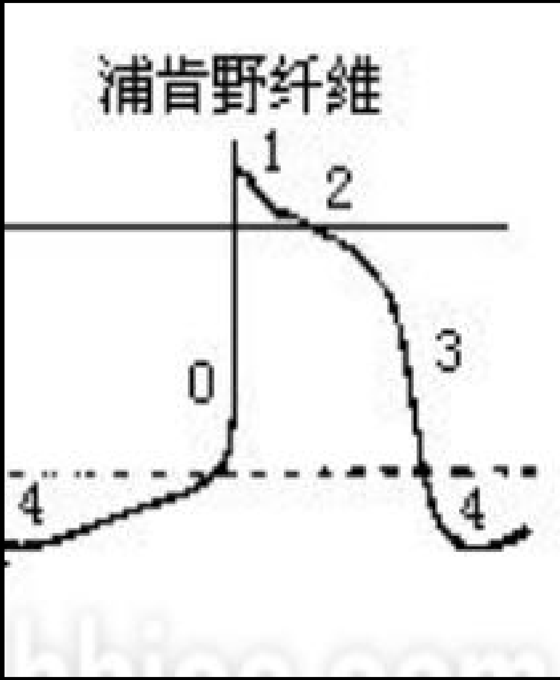
与心室肌细胞相比

✓最大复极电位更负（I_{k1}通道密度高）　比-90 mV更负

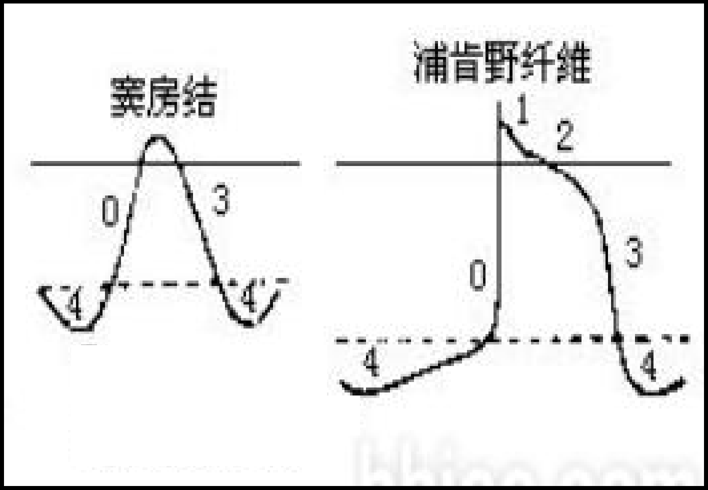
✓0期去极化速度更快 Fast response cell

✓1期更明显

✓4期自动去极化



与窦房结P细胞相比



✓I_f在4期自动去极化中起主要作用

- 3期复极化至-50mv时，I_k通道开始关闭，I_k电流逐渐减小，
- 同时，I_f通道开始缓慢激活开放，到-100mv时充分激活

4期自动去极化速度：
Purkinje: 0.02 V/s
SAN P cell: 0.1 V/s

✓4期自动化去极速度很慢 (I_f通道密度过低)
→动作电位时程最长的心肌细胞

II. The Physiological Properties of Cardiac Cells

工作细胞
心房肌
心室肌
快反应细胞
浦肯野
自律细胞
窦房结
房室结
慢反应细胞

- ✓ Excitability 兴奋性
- ✓ Conductivity 传导性
- ✓ Autorhythmicity 自律性
- ✓ Contractility 收缩性

–电生理特性

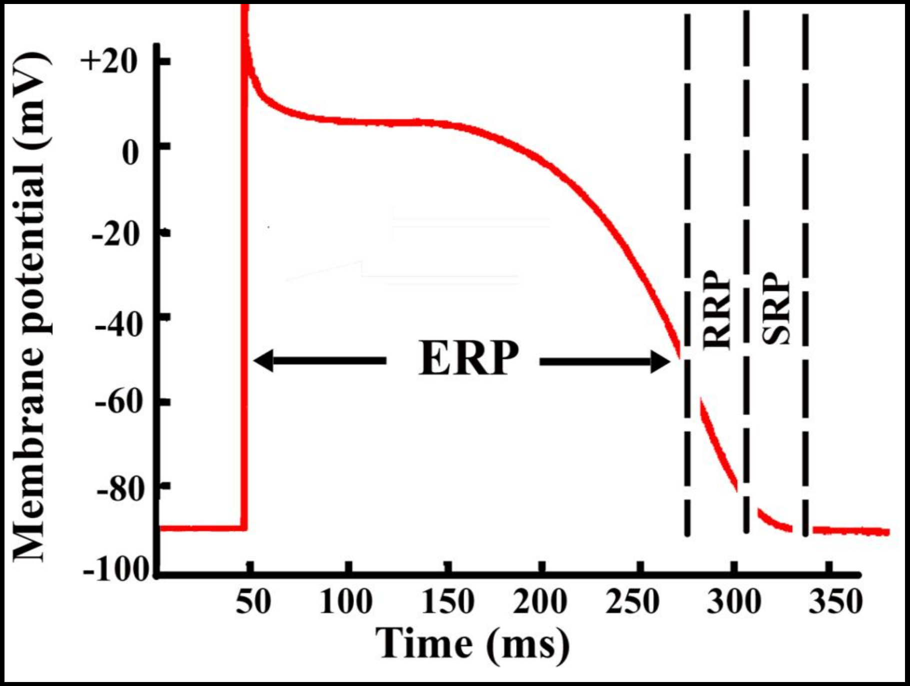
–机械特性

兴奋-收缩脱耦联（excitation-contraction decoupling）

1. Excitability (兴奋性)

以心室肌细胞为例

- **Effective refractory period (ERP，有效不应期): 0期~ -60 mv ,**
- **Relative refractory period (RRP，相对不应期): -60 ~ -80 mv**
- **Supernormal period (超常期): -80 ~ -90mv**

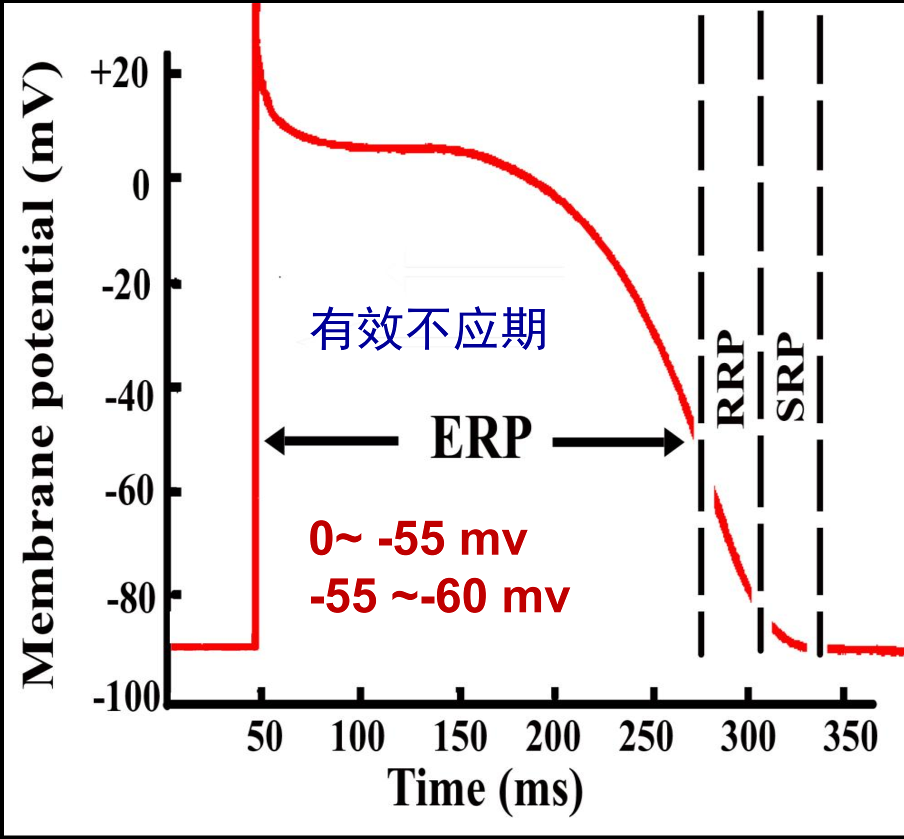


✓The refractory period of cardiac cells is long (~200 ms). compared to ~2 ms in neurons and skeletal muscle fibers.

Effective refractory period (有效不应期)

- **Absolute refractory period (绝对不应期):**
  - ✓ phase 0~ -55 mv
  - ✓ 无论多强刺激都不会引起去极化反应
  - ✓ 钠通道完全失活

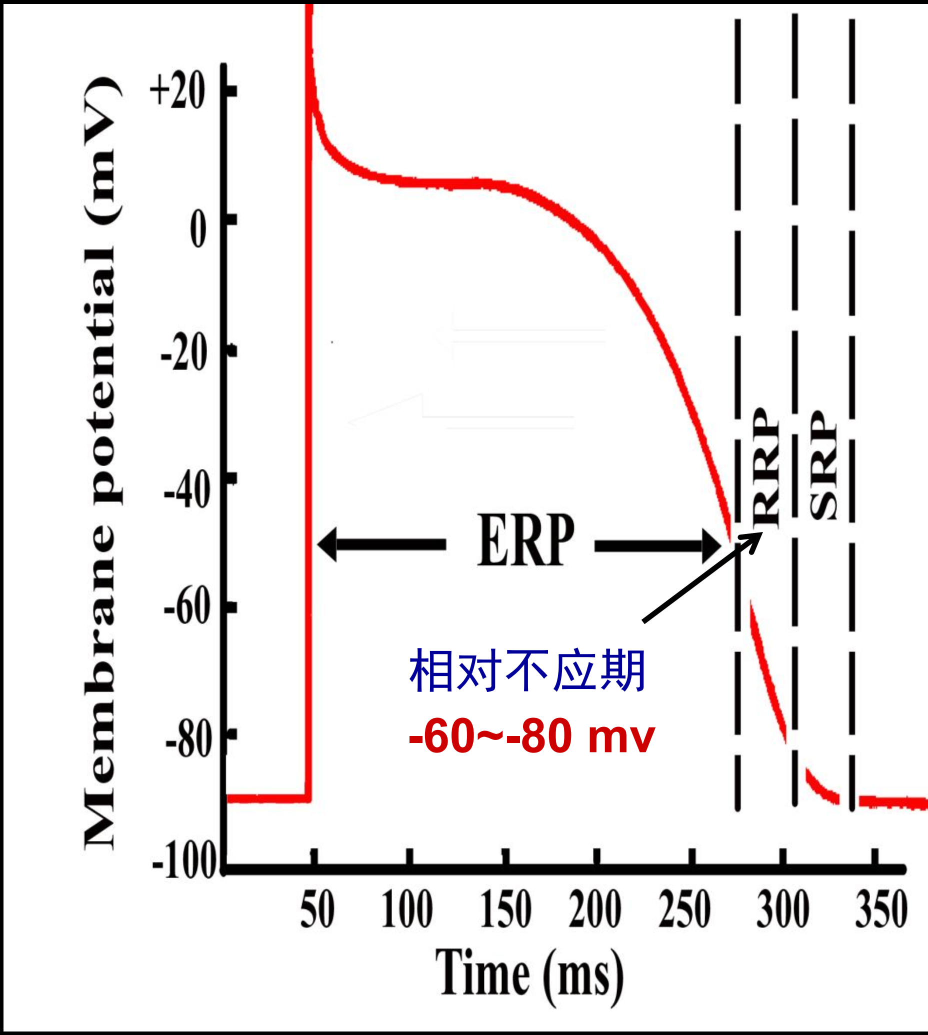
- **Local response period (局部反应期) :**
  - ✓ phase -55~ -60 mv
  - ✓ 阈上刺激可引起局部反应, 但不产生AP
  - ✓ 钠通道仅有少量复活



Relative refractory period (相对不应期)

-60 ~ -80 mv

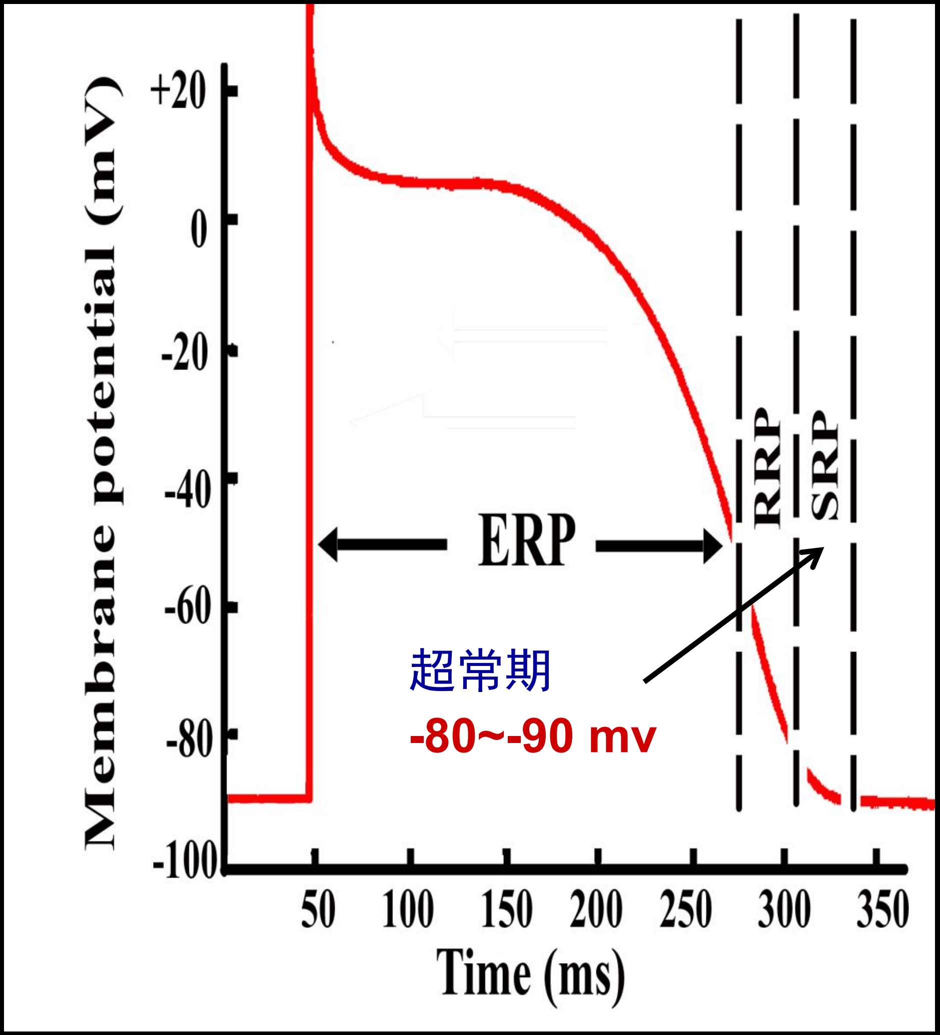
- **阈上刺激可引起可产生AP**
- **相当数量的Na^+通道恢复到备用状态，阈刺激不足以引起兴奋，加强刺激可引起新的AP**

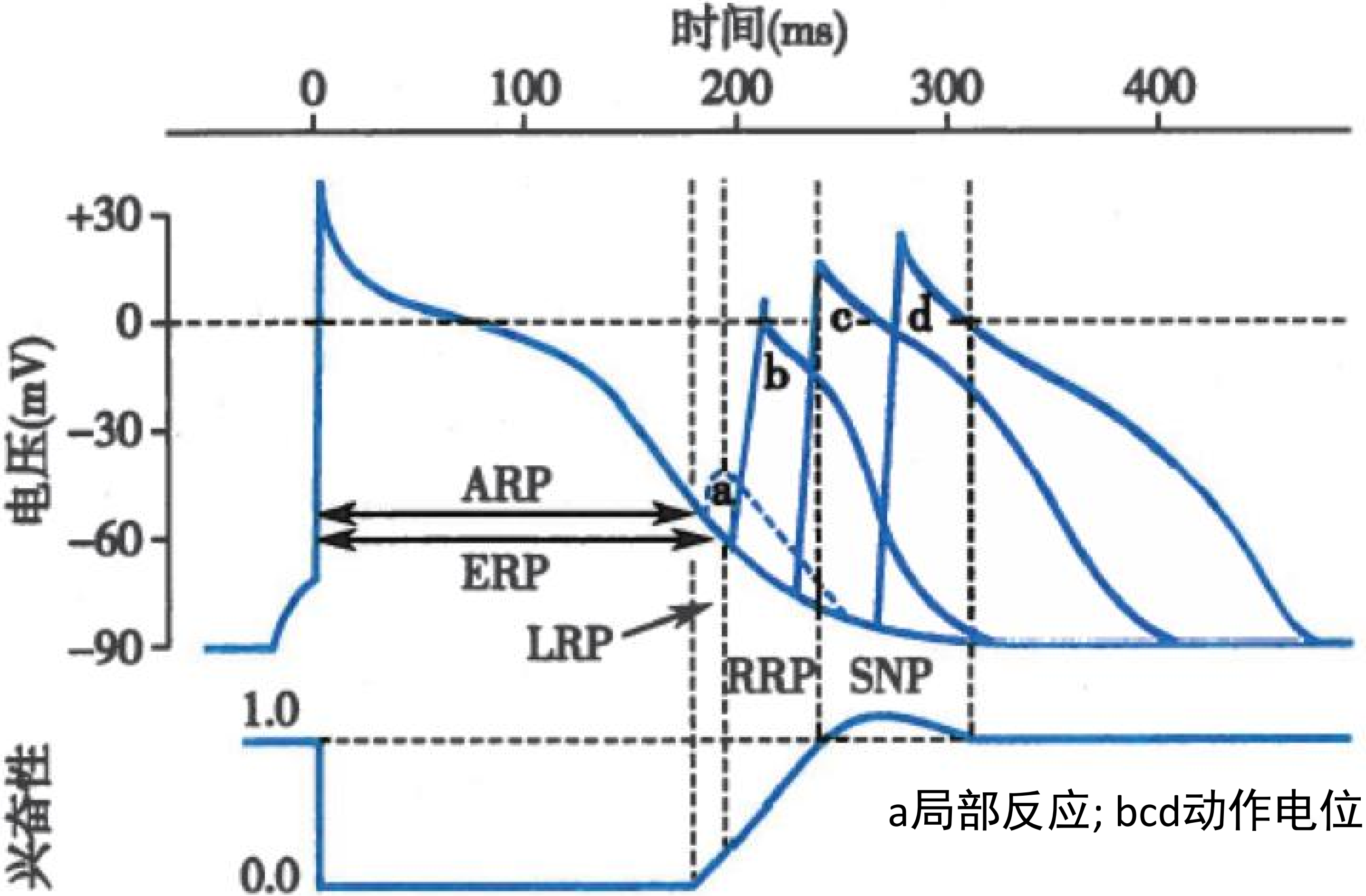


Supernormal period (超常期)

-80 ~ -90 mv

- **给予阈下刺激可引起AP**
- **大多数Na^+通道恢复到备用状态，且此期内膜电位距阈电位（-70 mV）较近，易兴奋**





在**相对不应期和超常期**，膜电位低于静息电位水平

Na^+通道开放的速率和数量都低

0期去极化的**速度和幅度**都低于正常。

Factors affecting excitability of cardiac cells

✓ The level of resting potential（静息/最大复极电位）

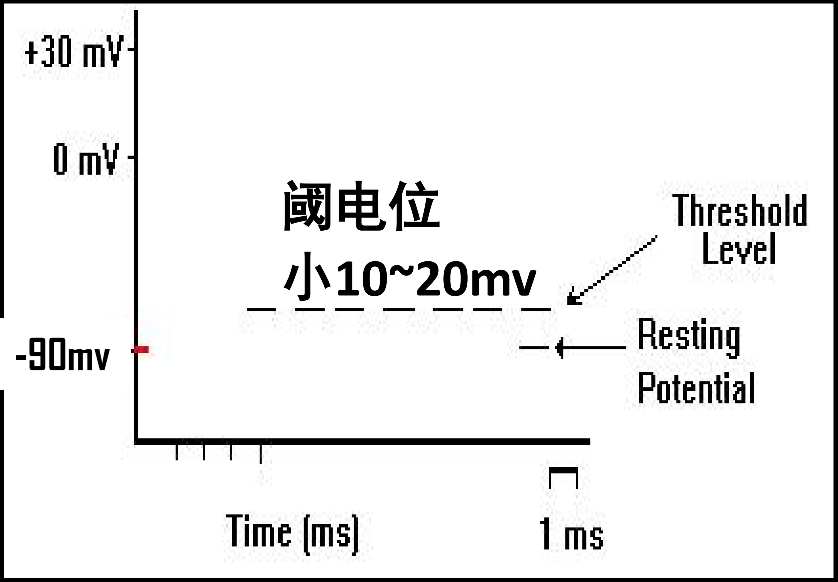
细胞外K+轻度升高，膜电位轻度去极，兴奋性升高

细胞外K+明显升高，膜电位显著减小，钠通道失活，兴奋性降低

✓ The level of threshold membrane potential（阈电位）

低血钙促进钠内流，阈电位下移，兴奋性升高

奎尼丁抑制钠内流，阈电位上移，兴奋性降低



***阈电位：能触发AP的膜电位临界值**

Factors affecting excitability of cardiac cell

✓ **State of Na^+ /Ca^{2+} channel（0期去极化离子通道性状）**

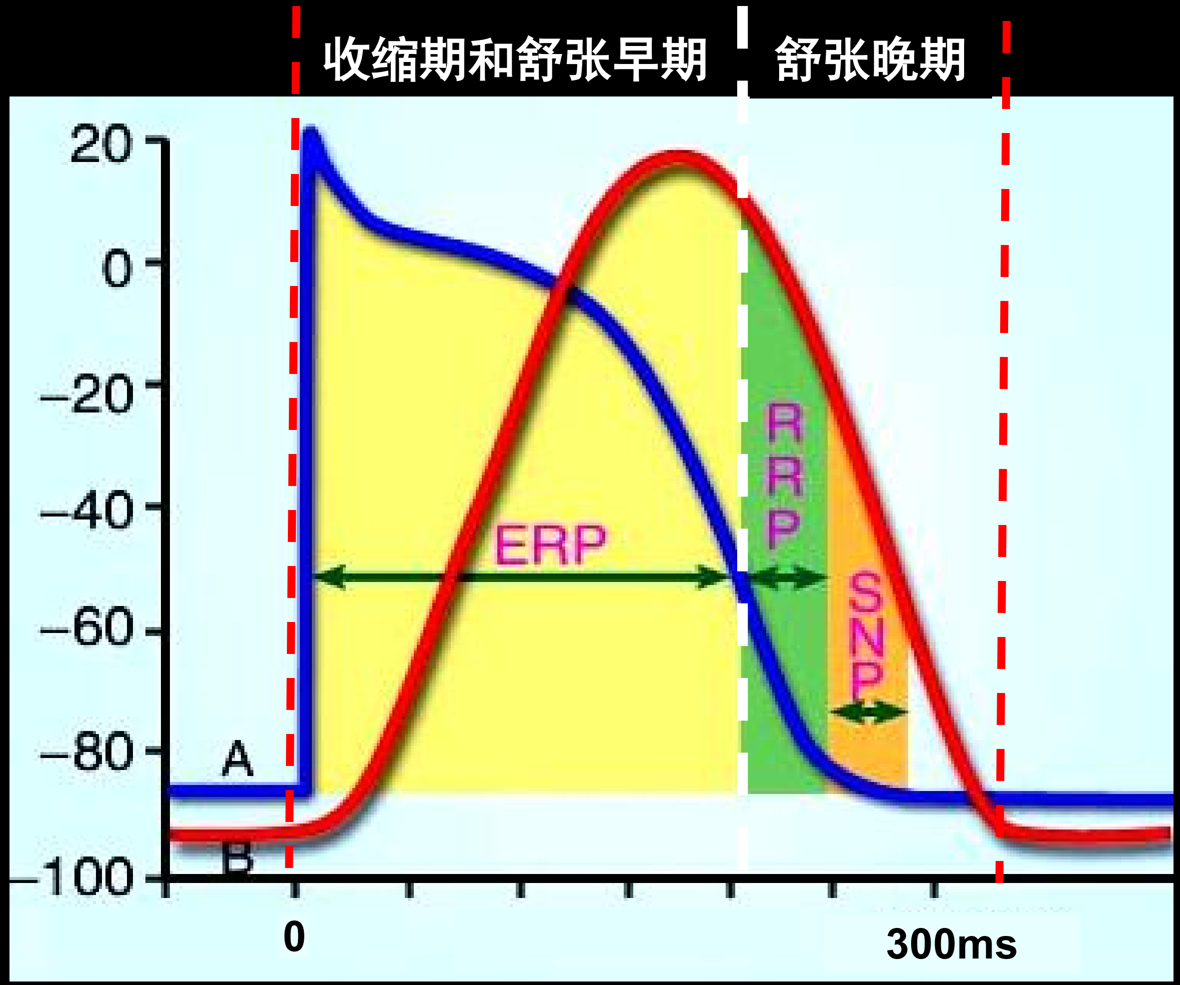
通道是否处于静息（备用）状态是心肌细胞是否具有兴奋性的前提

快反应细胞：Na^+通道

慢反应细胞：Ca^{2+}通道

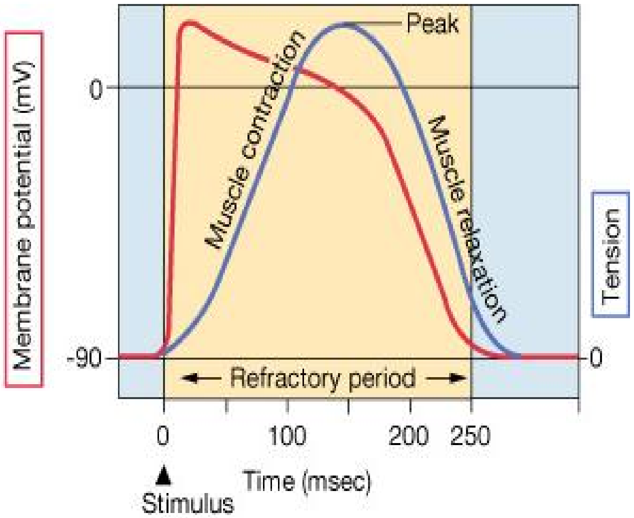
Na^+通道：	完全备用 →	失　活 →	刚复活 →	渐复活 →	基本备用
	‖	‖	‖	‖	‖
	产生AP	绝对不应期	局部反应期	相对不应期	超常期
	‖	‖	‖	‖	‖
兴奋性：	兴奋性正常	兴奋性无		兴奋性低于正常	兴奋性高于正常

Relationship between excitability & myocardial contraction

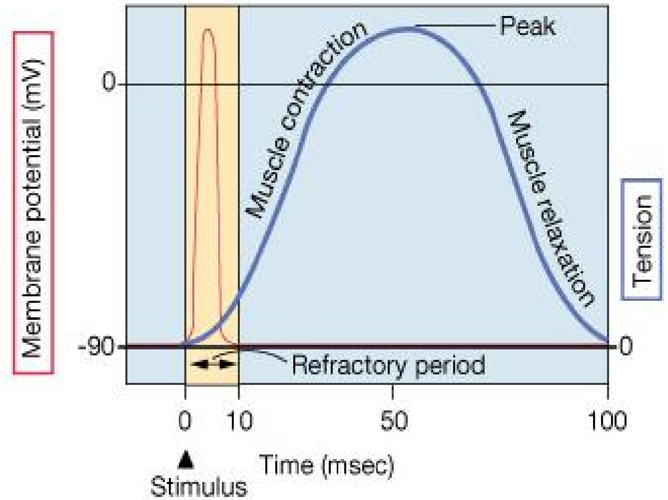


心肌不发生完全强直收缩，始终收缩和舒张交替，保证正常泵血

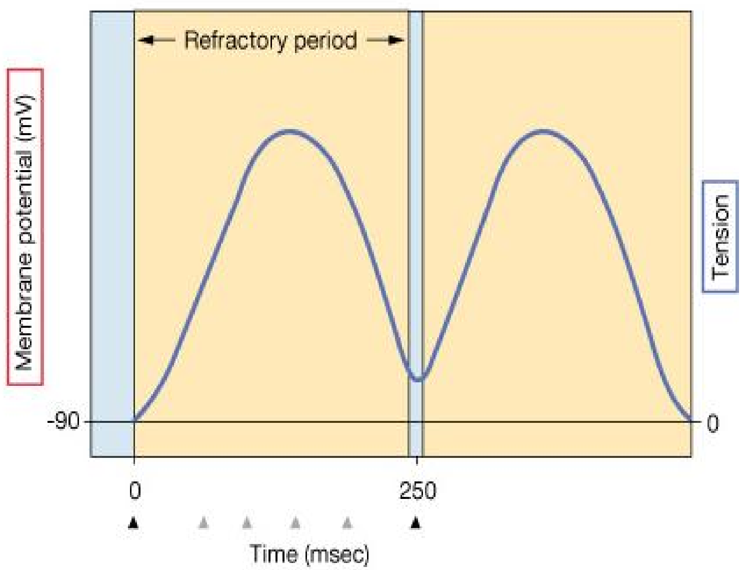
(c) Cardiac muscle fiber: The refractory period lasts almost as long as the entire muscle twitch.



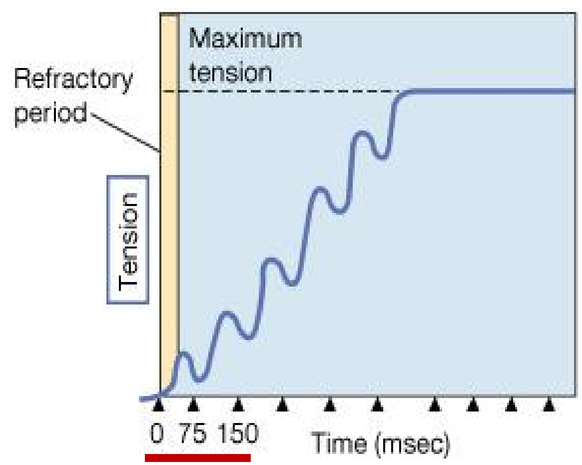
(a) Skeletal muscle fast-twitch fiber: The refractory period (yellow) is very short compared with the amount of time required for the development of tension.



(d) Long refractory period in a cardiac muscle prevents tetanus.



(b) Skeletal muscles that are stimulated repeatedly will exhibit summation and tetanus (action potentials not shown.)



每75ms一个刺激
不完全强直收缩

Extra systole/premature systole（期前收缩）:如果在心室肌的有效不应期后，下一次窦房结兴奋到达前，心室受到一次外来刺激，可提前产生一次兴奋和收缩。

Compensatory pause（代偿间歇）:

心房和心室在一次期前收缩之后，出现较长时程的舒张期

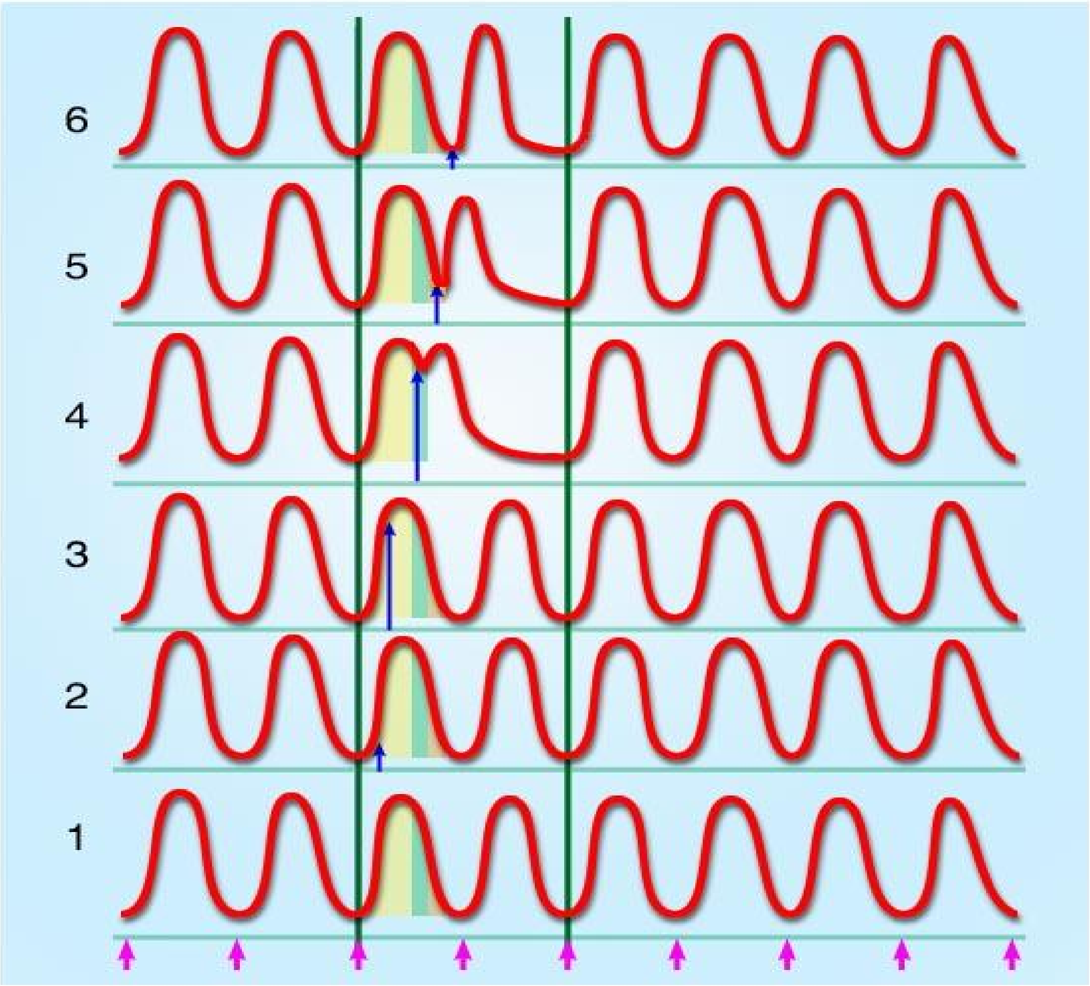
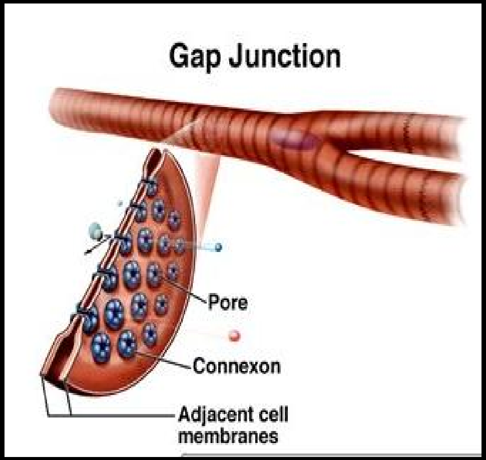


图 — 期前收缩与代偿间歇

2. Conductivity (传导性)

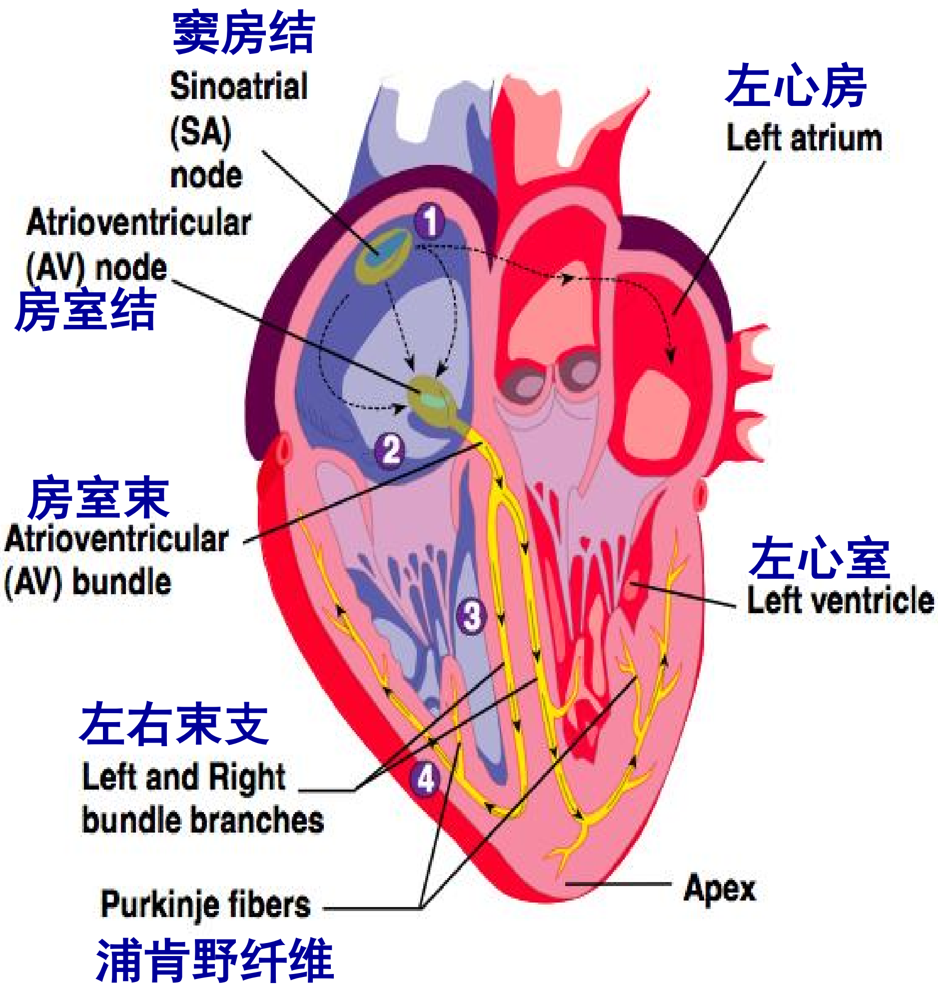
心肌细胞具有传导兴奋的能力或特性，称为~

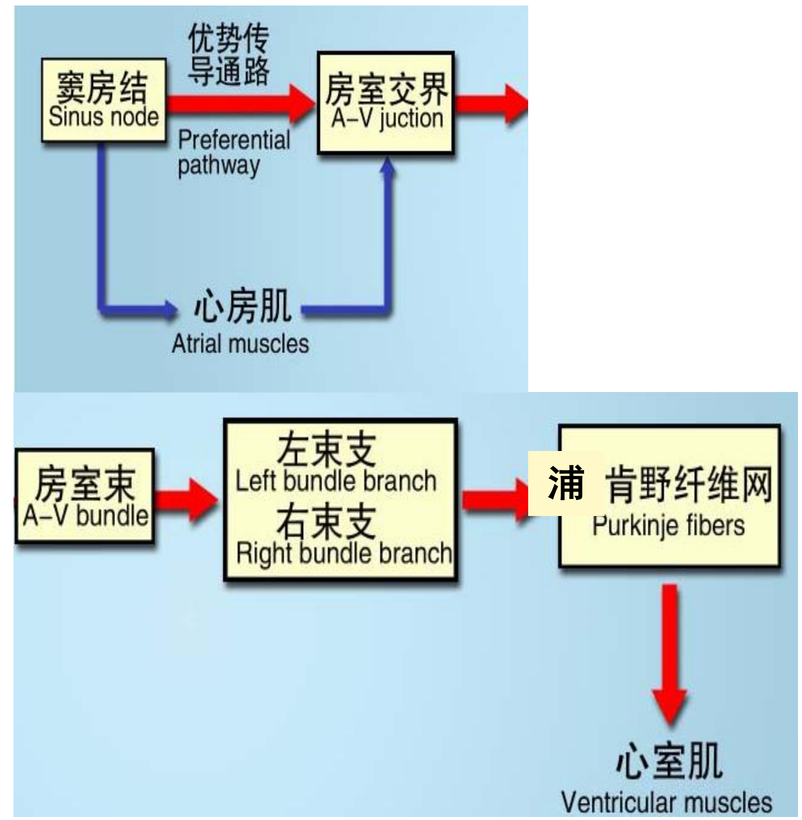
- ✓ 细胞内传导
- ✓ 细胞间传导 (闰盘-缝隙连接)



- ✓ 各类心肌细胞都有传导性
- ✓ 分化出特殊的传导系统

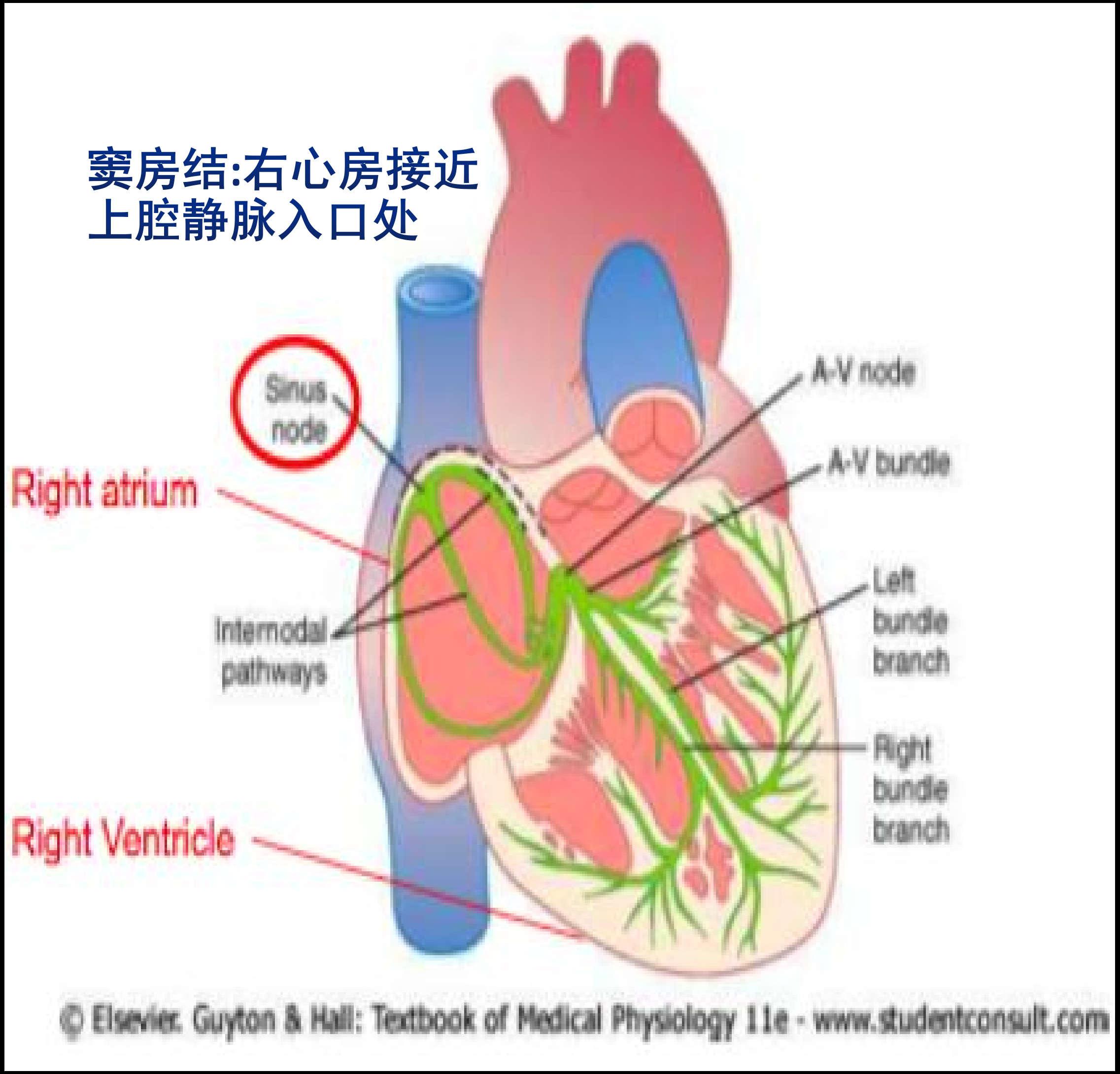
The conduction system of the heart



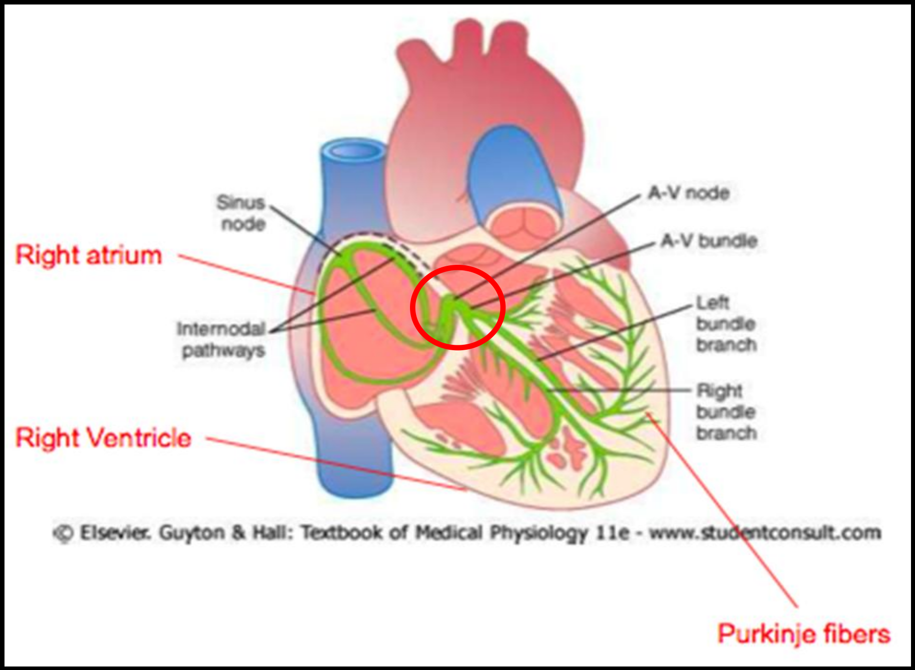


Sinoatrial Node (SAN): 窦房结

- ✓ 心脏起搏点
- ✓ 无收缩性
- ✓ 直接传递兴奋到心房肌纤维
- ✓ 优势传导通路传递兴奋到房室结



A-V Node (AVN): 房室结/房室交界

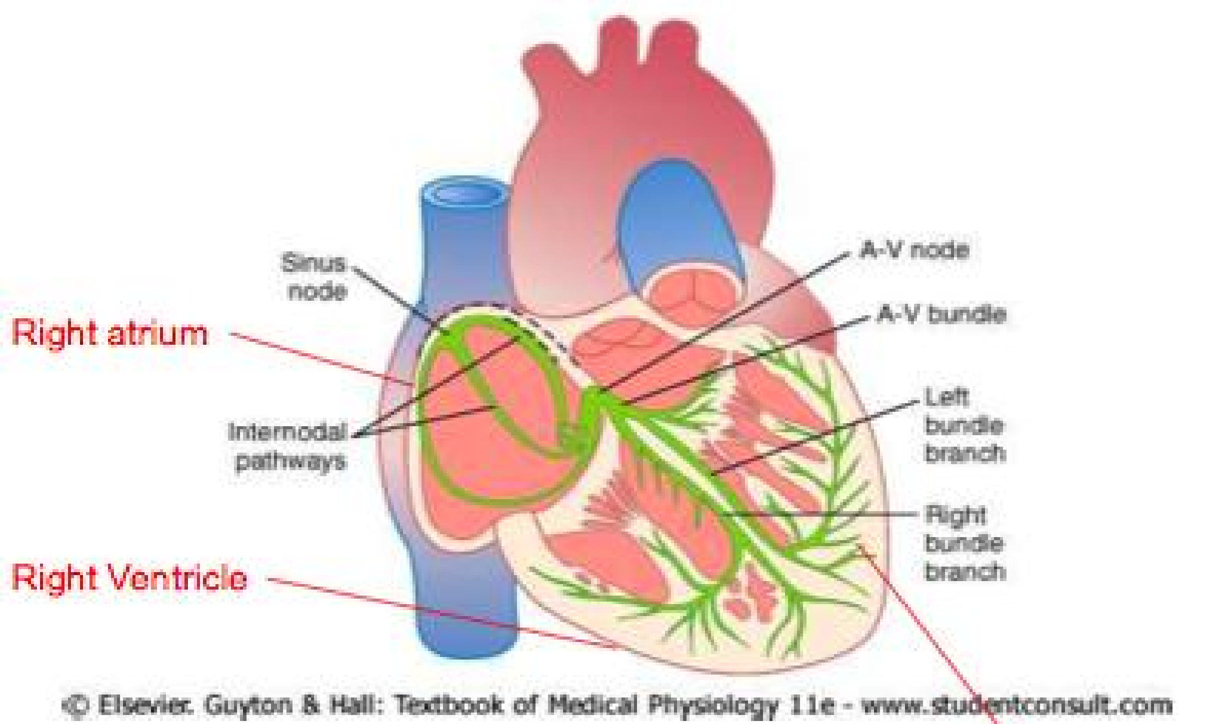


1.直径小
2. 闰盘处的缝隙连接少
3.分化程度低，传导兴奋能力低

Atrioventricular delay (房-室延搁) → 房室传导阻滞

房室结区的传导速度缓慢，且房室结是兴奋由心房传向心室的唯一通道，兴奋经过此处将出现一个时间延搁

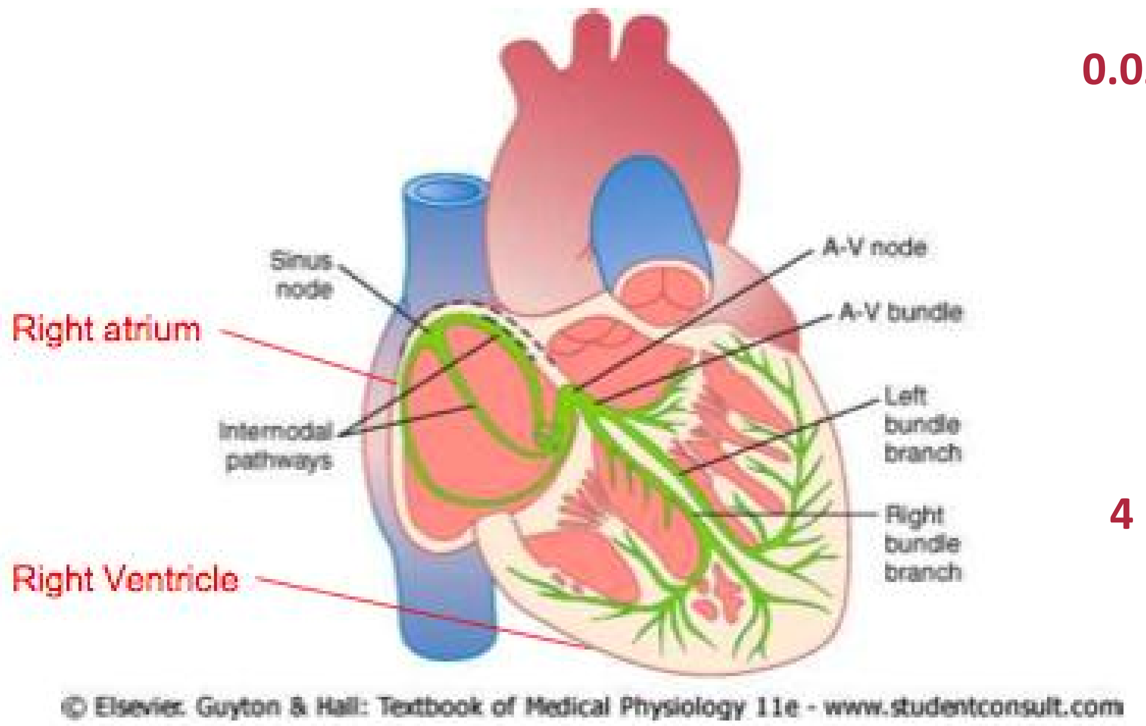
Purkinje Fibers



1.直径大
2. 闰盘处的缝隙连接多

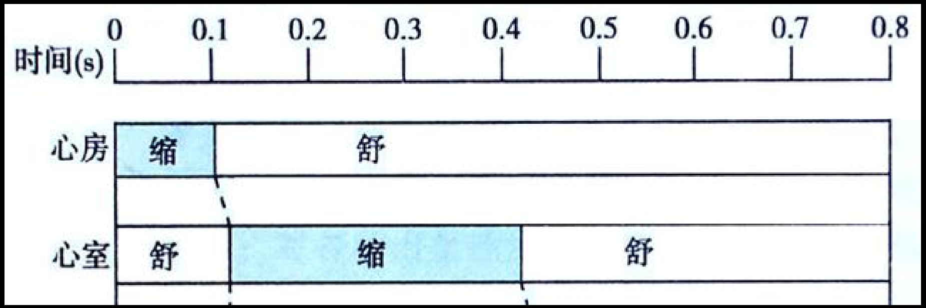
兴奋传导速度最快，迅速传到心室肌

Summary of Path of Propagation



	SA node		**Pacing (sets HR)**
	⬇		
0.02 m/s	AV node		**0.15 s**
	⬇		
	AV bundle		
	⬇		
4 m/s	Purkinje fibers		
	⬇		
	Ventricles		
	⬇		
	Outer surface of heart		

Characteristics of conduction in heart



- ✓ Delay in transmission at the A-V node (~100 ms) – sequence of the atrial and ventricular contraction – 生理意义
- ✓ Rapid transmission of impulses in the Purkinje system – synchronize contraction of entire ventricles –生理意义

Factors determining conductivity

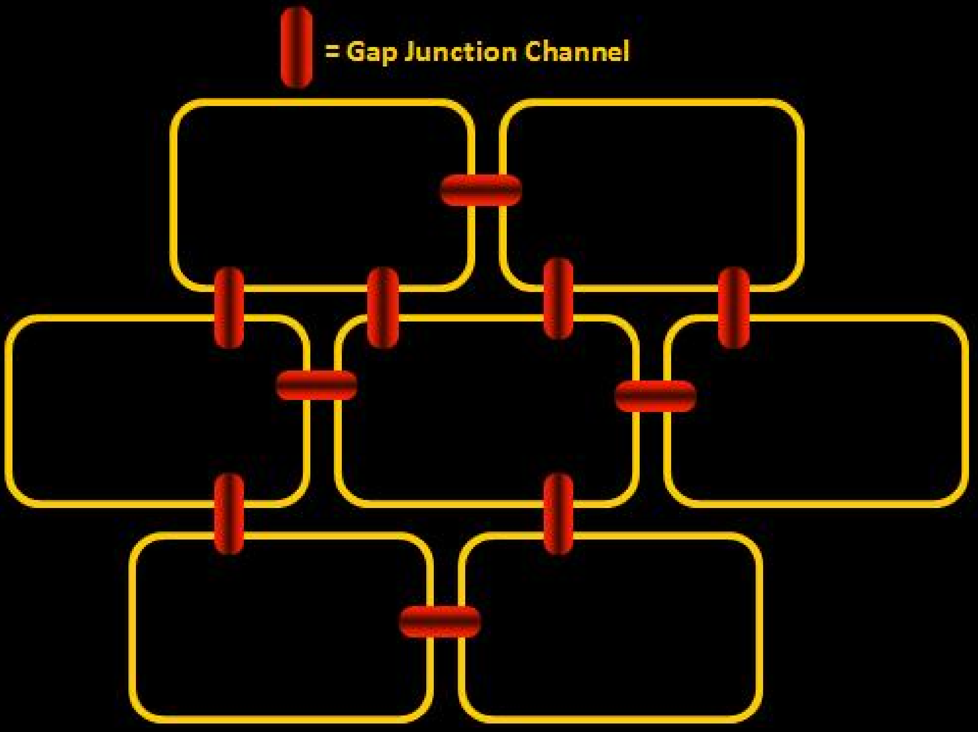
- ✓ Anatomical factors
- ✓ Physiological factors

Anatomical factors affecting conductivity

✓ **Diameter of the cardiac cell ↑**

→ **conductive resistance ↓**

→ **conductivity↑**



✓ **Gap junction**

—心肌缺血时可关闭

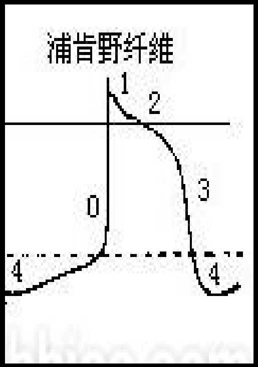
✓ **Differentiation**

Physiological factors affecting conductivity

1. Amplitude & speed of phase 0 (0期去极化的速度和幅度)

最重要因素

0 期除极速度快	0 期除极幅度大
↓	↓
临近未兴奋膜局部电流形成快	(兴奋和未兴奋部位之间) 电位差大
↓	↓
达到阈电位快	局部电流强，扩布距离大，使更远处兴奋
↓	↓
兴奋传导快	兴奋传导快



Purkinje细胞
0期去极化速度快！

奎尼丁抑制 I_{Na}
(I类抗心律失常药)
0期去极化速度慢！

Physiological factors affecting conductivity

2. Membrane potential

Affecting amplitude & speed of phase 0

期前兴奋传导减慢！

受刺激前的膜电位水平

→Na^+通道性状

→膜去极化后通道开放的速度与数量

→膜0期去极化的速度和幅度

膜电位较小

↓

Na^+通道不在最佳状态

↓

0期去极化的速度和幅度↓

↓

传导减慢

Physiological factors

3. The excitability of adjacent membrane

邻旁未兴奋区兴奋性降低

↓

达阈电位所需时间长

↓

传导速度减慢

兴奋

↓

邻旁未兴奋区有效不应期

↓

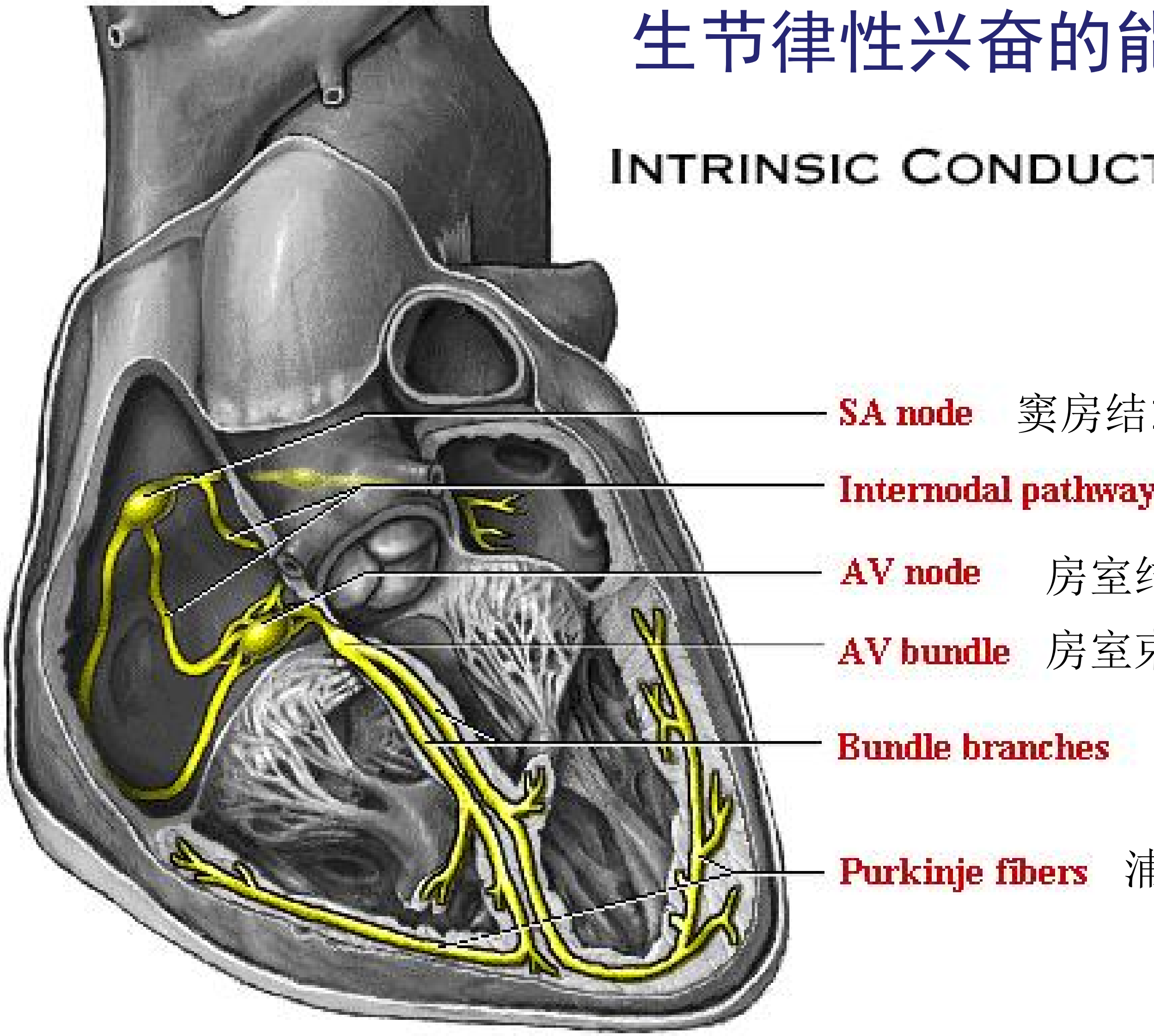
不能引起兴奋

↓

传导阻滞

3. Autorhythmicity (自动节律性)

心肌在无外来刺激条件下能自动产生节律性兴奋的能力或特性。

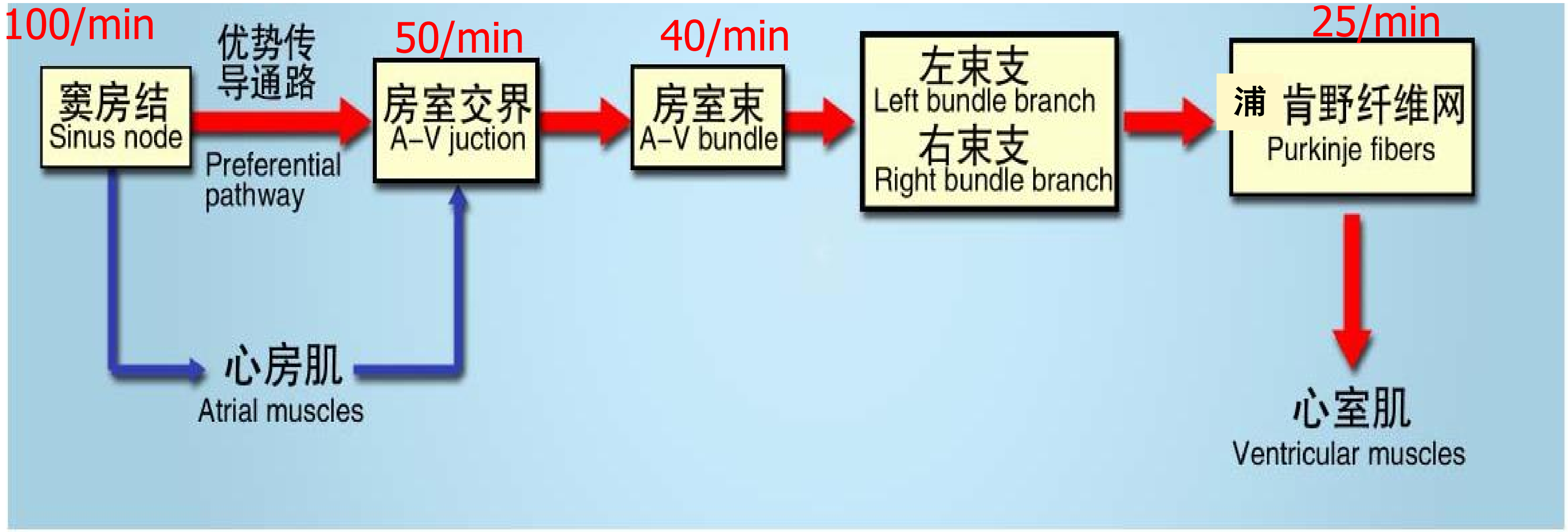


窦房结100/min (70/min)

房室结（交界）50/min

房室束40/min

浦肯野细胞纤维25/min



Normal pacemaker 正常起搏点：窦房结

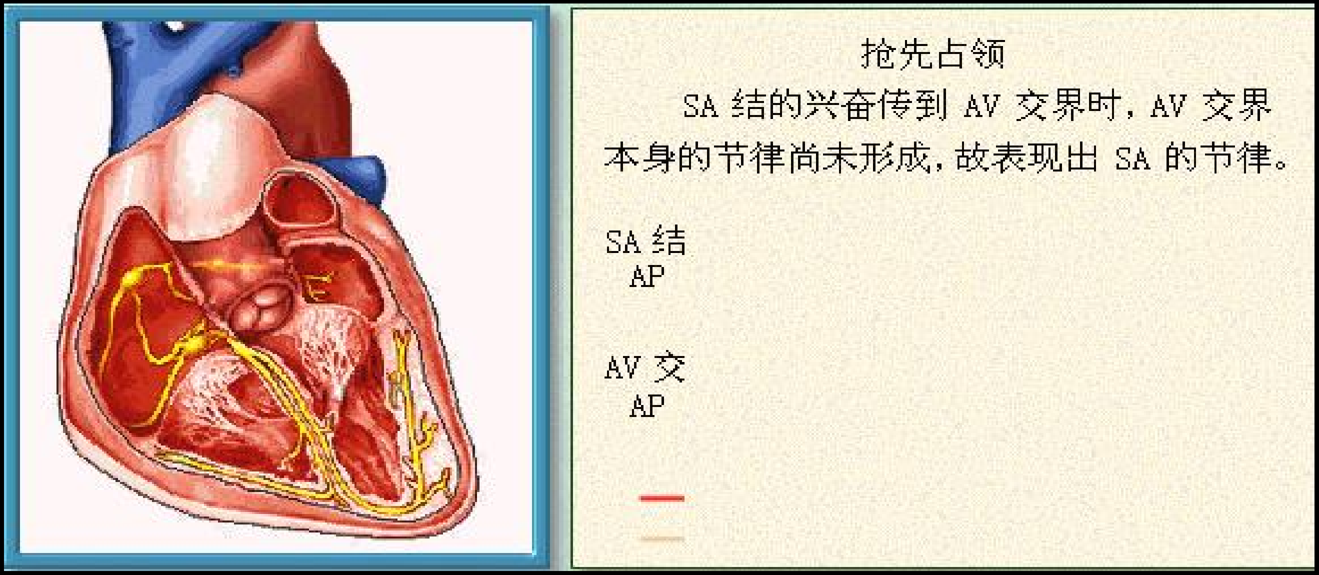
Sinus rhythm 窦性节律：由窦房结起搏而形成的心脏节律。

Latent pacemaker 潜在起搏点：正常情况下，心脏其他自律性组织仅起到兴奋传导的作用，而不表现出其自身的节律性。

Ectopic pacemaker 异位起搏点：在某些病理情况下，正常起搏点起搏功能障碍或者传导发生障碍，或潜在起搏点的自律性异常增高超过窦房结时，可代替窦房结产生可传播的兴奋而控制心脏的活动，此时异常的起搏部位称异位起搏点。

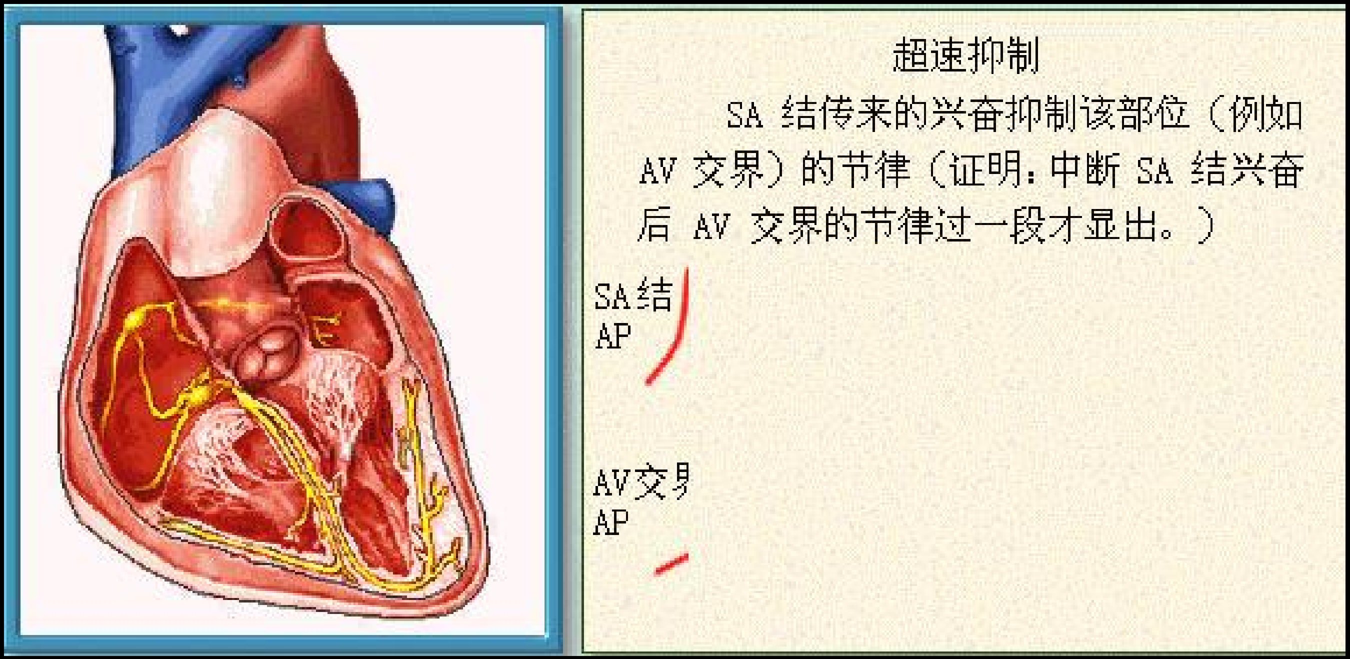
➢Capture effects (抢先占领)

由于窦房结 的自律性高于其他潜在起搏点，因此潜在起搏点在其自身去极化达到阈电位之前，由窦房结传来的兴奋已将其激活而产生动作电位，从而控制心脏的节律活动。



➢ Overdrive suppression (超速驱动压抑)

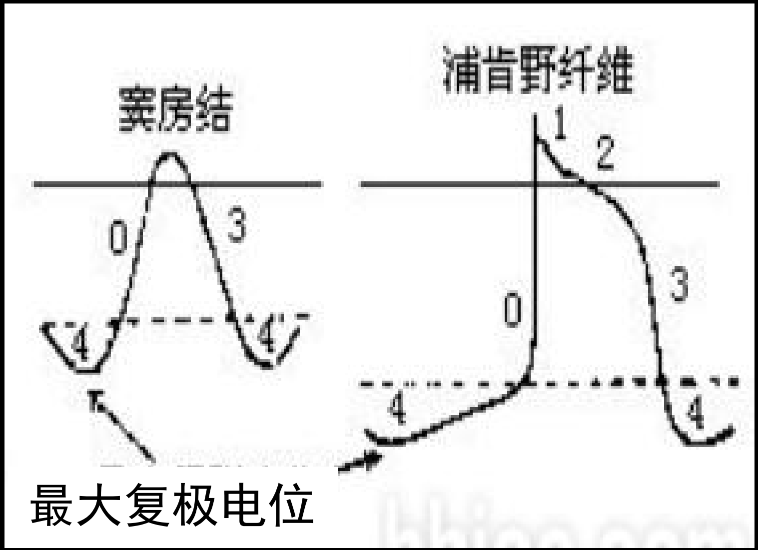
在外来超速驱动刺激停止后，自律细胞不能立即呈现其固有的自律性活动，需经过一段静止期后才逐渐回复其自身的自律性活动。



Factors affecting autorhythmicity

- Speed of phase 4 **最重要因素**

  4期自动去极化速度

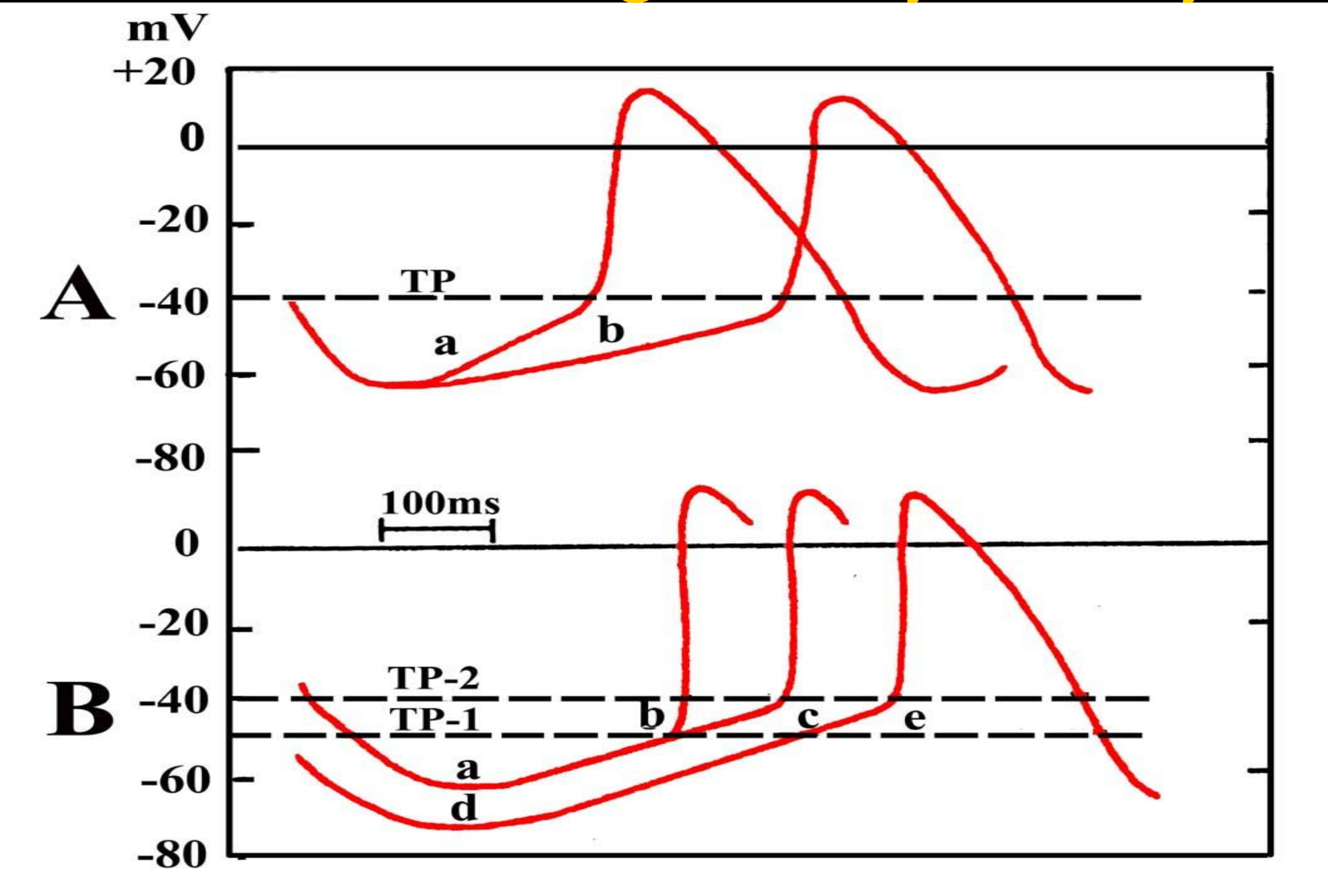


- Level of maximal repolarization potential (MRP)

  最大复极电位水平

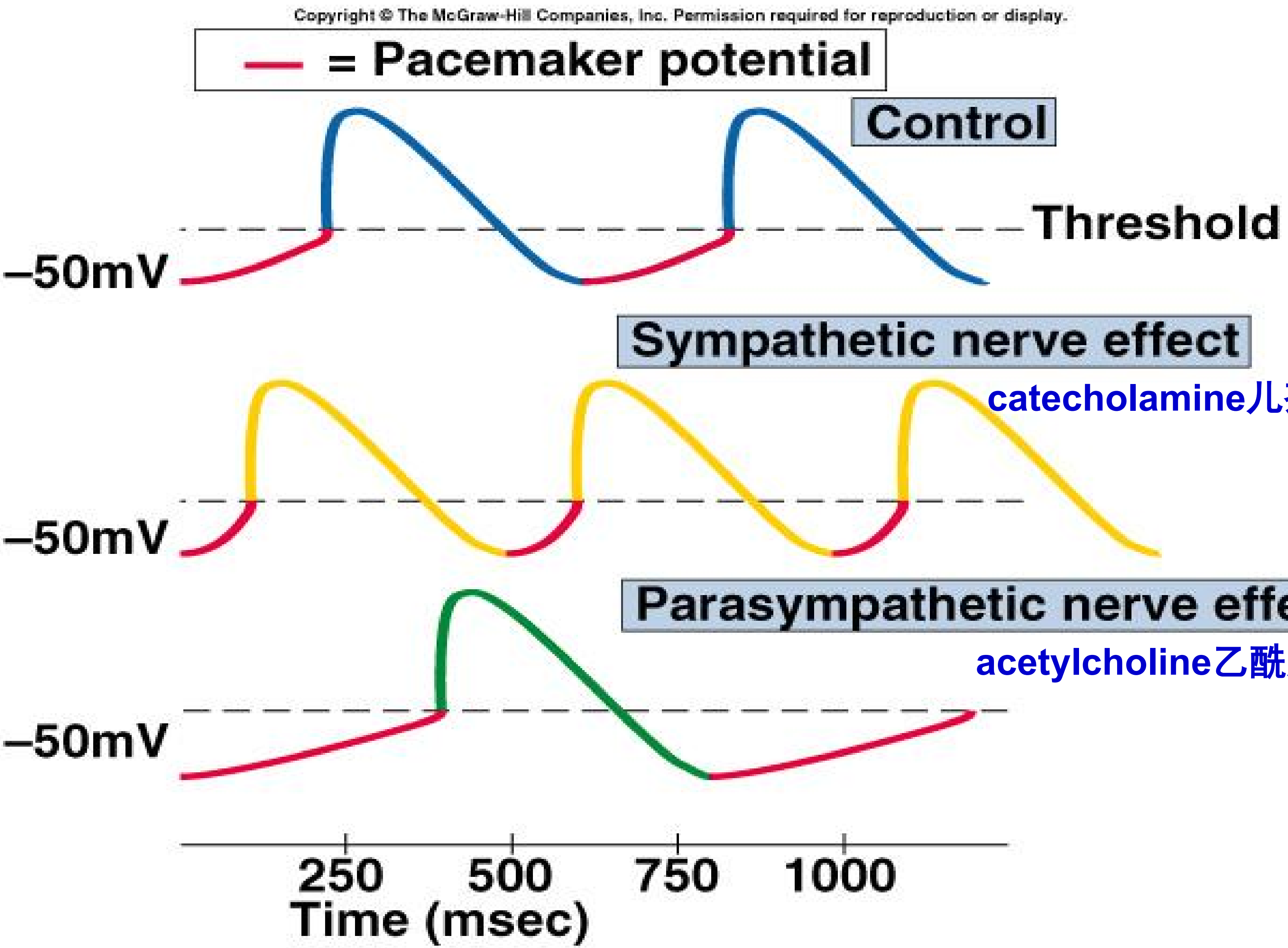
- Level of threshold potential **阈电位水平**

Factors affecting autorhythmicity



A . 4期自动去极化速率由a减到b时，自律性降低
B.阈电位由TP-1(ab)到TP-2(ac)或最大复极电位由a(ac)超极化到d(de), 自律性降低

Modulation Pacemaker Activity.



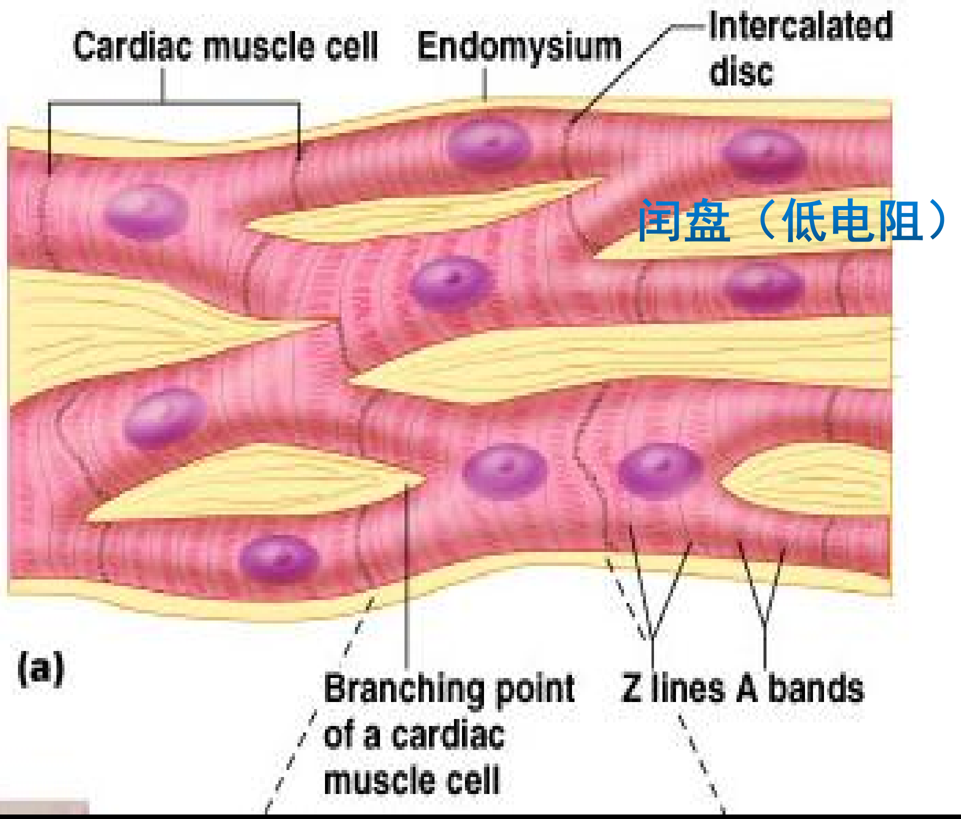
catecholamine儿茶酚胺

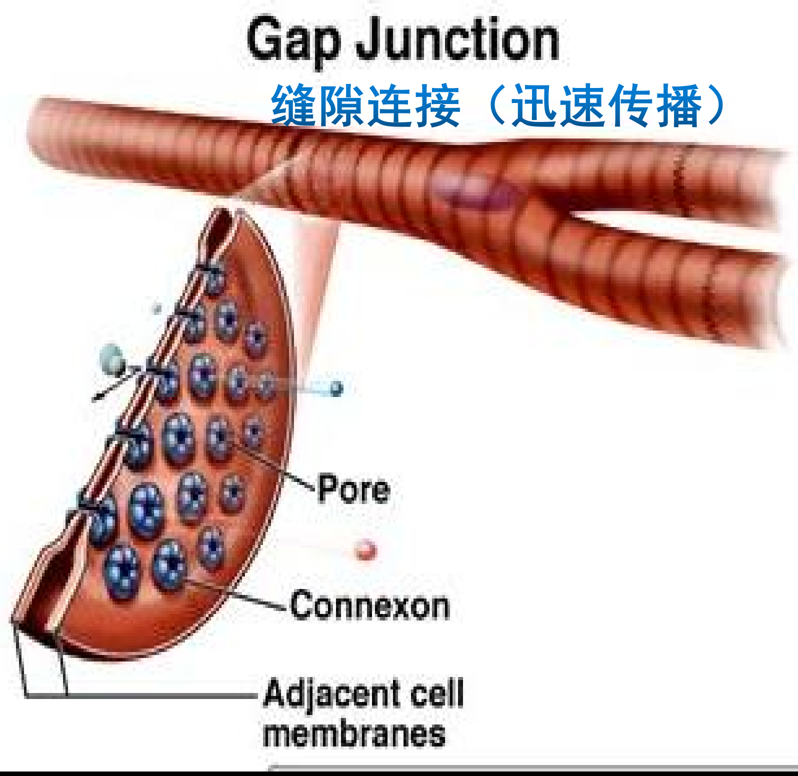
acetylcholine乙酰胆碱

4. Contractility (收缩性)

(1) Synchronized contraction “all and none”

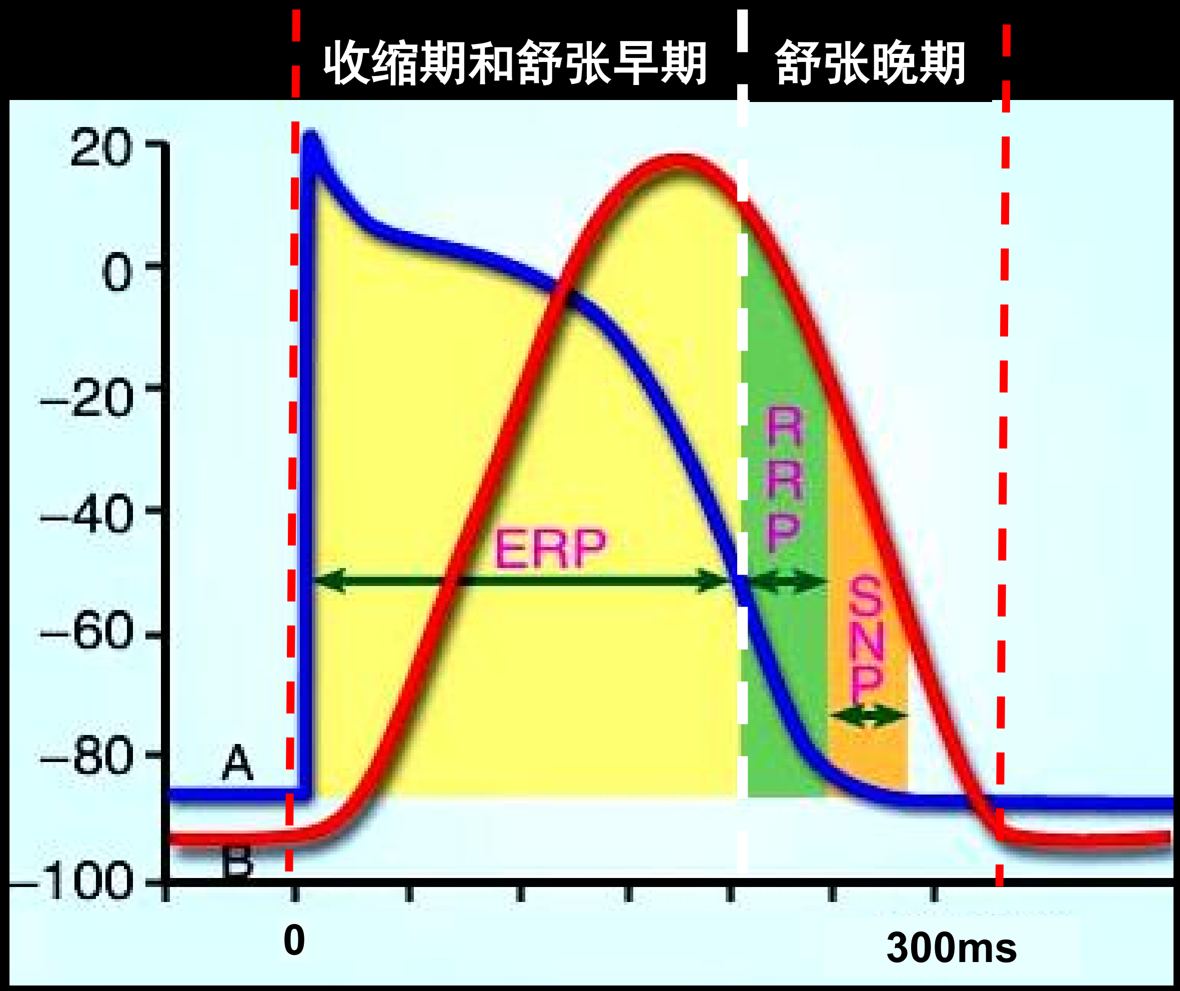
Two functional syncytiums(合胞体): Atrium/Ventricle





(2) No tetanus

Cardiac muscles must relax between contractions so the ventricle can fill with blood



(3) Extracellular Ca^{2+} dependent in Excitation-Contraction Coupling and Relaxation

Calcium-Induced Calcium Release (CICR)
钙诱导钙释放